

Topography, Architecture, and Connections of Somatosensory Cortex in Opossums: Evidence for Five Somatosensory Areas

PAMELA D. BECK, MARCIE W. POSPICHAL, AND JON H. KAAS
Department of Psychology, Vanderbilt University, Nashville, Tennessee 37240

ABSTRACT

Microelectrode maps of somatosensory inputs were related to cortical architecture and patterns of cortical connections to provide evidence for five subdivisions of the somatosensory or sensorimotor cortex in North American opossums (*Didelphis marsupialis*). Microelectrode recordings revealed three systematic representations of the body surface. A large mediolaterally oriented representation was identified as the primary somatosensory area (S1) by its relative position, somatotopy, architecture, and connections. S1 represented the hindlimb, trunk, forelimb, and face in a mediolateral sequence. Two additional representations of cutaneous receptors were found caudolateral to S1, each with face representations adjacent to the border of lateral S1 and other body-part representations progressing more caudally toward the auditory cortex. We identified the more dorsal field as the second somatosensory area (S2) and the more ventral field as the parietal ventral area (PV). Tracers injected into S1 labeled neurons and terminals in architectonically distinct fields rostral and caudal to S1, the somatosensory caudal area (SC) and the somatosensory rostral area (SR). Movements could be evoked by microstimulation from sites scattered over S1, SR, and the frontal cortex, but thresholds were high and uncharacteristic of motor cortex. S2 and PV merged caudally with the cortex responsive to auditory stimuli, possibly A1, and neurons in some caudal recording sites in PV were activated by both auditory and cutaneous stimuli. Primary (V1) and secondary (V2) visual areas were also identified by microelectrode mapping, architecture, and connections. In addition, at least part of the cortex between V2 and the somatosensory cortex had visual connections. Thus, most of the dorsolateral cortex of opossums appears to be somatosensory, auditory, or visual. © 1996 Wiley-Liss, Inc.

Indexing terms: marsupial, motor cortex, visual cortex, auditory cortex, brain evolution

The present experiments are part of an ongoing effort to determine features of cortical organization likely to have been present in early mammals (see Kaas et al., 1970; Kaas, 1987). By 100 million years ago (MYA), mammals had diverged into three major surviving lines: marsupials or Metatheria, monotremes or Prototheria, and placental mammals or Eutheria (Rowe, 1993). By identifying features shared by living representatives of these groups, we can infer characteristics present in their last common ancestor (Wiley, 1981). Thus, it is important to investigate cortical organization in all three groups. A complication, however, is that brains are highly varied in fissure patterns, general proportions, and overall size across members of each group (e.g., see Welker, 1990), and if they vary as much in other features, the task of determining common features would be extensive. Fortunately, studies of endocasts of fossilized skulls (e.g., Jerison, 1990) indicate that mammals evolved from reptiles with rather small brains (in proportion to their bodies) and that mammals retained small brains until

50–60 MYA. Mammals at that time had small smooth brains with little neocortex, and the great variety in brain shapes has emerged relatively recently. Fortunately, some extant mammals in all three groups have brains that closely resemble those of early mammals, at least in superficial appearance, and it is in these species that it is most logical to investigate brains experimentally to determine the probable organization of the brains of the first mammals.

Because opossum brains are primitive in appearance and represent one of the three major branches of mammalian evolution, opossums have been studied extensively by comparative neurobiologists (for reviews, see Johnson, 1990; Rowe, 1990). Early studies led to important observations and conclusions, but interest peaked before many of the

Accepted August 18, 1995.

Address reprint requests to Jon H. Kaas, Ph.D., Department of Psychology, Vanderbilt University, 301 Wilson Hall, 111 21st Avenue South, Nashville, TN 37240.

presently available experimental methods were in common use, and thus uncertainties about cortical organization remain.

Several early influential studies have provided evidence that cortical organization in opossums is less complex than in most mammals. Most notably, Lende (1963a) presented evidence that the primary sensory (S1) and motor (M1) fields of eutherian mammals are "amalgamated" in opossums, to constitute a single area (SM1) rather than two. In addition, a second somatosensory field (S2), just lateral to SM1, appeared to be within the auditory cortex. Thus, the neocortex in opossums appeared to contain fewer subdivisions of sensory cortex than that in eutherian mammals, and auditory and somatosensory fields were not completely differentiated from each other.

The conclusions of Lende (1963a) were based on recordings and stimulations with surface electrodes, methods that yield only crude localization, especially in small brains. Nevertheless, these conclusions have been supported in part by the results of subsequent studies. Microelectrode recording studies (Pubols et al., 1976) have provided additional evidence for an S1, albeit for a smaller S1 than that indicated by Lende (1963a), and for an S2 that is also responsive to auditory stimuli (Pubols, 1977). Furthermore, additional electrophysiological (Magalhães-Castro and Saraiva, 1972) and new anatomical (Donoghue and Ebner, 1981) studies have provided evidence of sensorimotor overlap in S1 or SM1. Thus, early studies suggested that opossums have no more than two sensorimotor fields and that one of these fields, S2, is not even completely differentiated from auditory cortex.

Although the results of more recent investigations are largely consistent with Lende's (1963a) original descriptions, there are reasons to reexamine the number of subdivisions of somatosensory or sensorimotor cortex in opossums. First, the smaller S1, as defined by the microelectrode recordings of Pubols et al. (1976), leaves caudal and rostral parts of parietal cortex (Gray, 1924) without known roles. These regions could be subdivisions of somatosensory or sensorimotor cortex. Second, reinvestigations of the region of cortex formerly designated as S2 in a number of mammals have revealed an additional somatosensory area. A third systematic representation of the body surface in cortex ventral to S2 and caudal to S1, the parietal ventral somatosensory area (PV), was first discovered in squirrels (Krubitzer et al., 1986). The parietal ventral area, in addition to S2, has now been described in rats (Li et al., 1990; Fabri and Burton, 1991), bats (Krubitzer and Calford, 1992), and both New World (Krubitzer and Kaas, 1990) and Old World (Krubitzer et al., 1995a) monkeys. Somatosensory representations that probably correspond to PV have also been described in cats (Clemo and Stein, 1982), moles (Catania and Kaas, 1995), and both platypuses and echidnas (Krubitzer et al., 1995b). Given the broad distribution of PV or a PV-like area in other mammals, it seems reasonable to investigate opossums further. Finally, there already is some evidence for a third somatosensory area in opossums. Pubols et al. (1976) described two adjoining representations of the mystacial vibrissae and the rhinarium, both of which they assigned to S1. The mystacial vibrissae and rhinarium were found to be represented a third time in more ventral cortex assigned to S2 (Pubols, 1977). Considered in the context of the more recent evidence for PV in other mammals, the results suggest the possibility of three representations of the body surface—

S1, S2, and PV—in opossums rather than an S1 with a double representation of part of the face and an S2.

Our experimental approach used the same methods that have been used to reveal somatosensory fields in other mammals (e.g., Krubitzer and Kaas, 1990). Thus, we used microelectrode recordings to determine the somatotopic organization of cortical sites responsive to tactile stimulation. In the same animals, we then related somatotopic organization to the cortical architecture in brain sections cut parallel to the cortical surface. In some of these opossums, we studied cortical connections by injecting tracers into physiologically identified locations in somatosensory cortex. We also determined cortical locations where microstimulation evoked movements and cortical sites where neurons were responsive to auditory or visual stimuli. Together, the results provide a comprehensive view of the extent and organization of sensorimotor cortex in relation to auditory and visual fields in opossums. A brief report of some of these results has appeared elsewhere (Beck and Kaas, 1993).

METHODS

Subdivisions and connections of the neocortex were investigated in 15 adult opossums (*Didelphis marsupialis virginiana*). Microelectrode multiunit mapping methods were used to investigate the somatotopic organization of the somatosensory cortex in 13 animals. Regions responsive to auditory stimuli also were noted in 12 of these animals to compare the location of the somatosensory to the auditory cortex. Results were later related to cortical architecture in sections from the flattened cortex of the experimental animals processed for Nissl substance, cytochrome oxidase, or myelin. Patterns of connections were studied following injections of anatomical tracers into physiologically identified sites in the somatosensory cortex in 4 of these animals. The cortex was electrically stimulated with microelectrodes in 3 of these animals to determine cortical regions where movements could be evoked. Finally, the relationship of the somatosensory to the visual cortex was explored by determining receptive fields for neurons and retinotopic organization over part of the primary visual cortex in two opossums and by placing an injection of anatomical tracer into the primary visual cortex in 1 opossum.

Electrophysiological recording and microstimulation

Two types of recording procedures were used. In four opossums, part of the somatosensory cortex was exposed while using aseptic procedures while the animals were deeply anesthetized with either 1–2% isoflurane or a combination of ketamine hydrochloride (130 mg/kg initial dose, i.m.) and acepromazine (4.5 mg/kg initial dose, i.m.). Maintenance doses of 10–20% of the original dose of anesthetic were given as needed to maintain a surgical level of anesthesia. Recordings were obtained at 1–10 cortical sites by using methods identical to more extensive recordings described below. Anatomical tracers were injected into one or more of these sites (see below). The opening in the bone was then closed with dental cement, the muscle and skin were sutured in place, and the animal was allowed to recover from anesthesia.

In other more extensive recording experiments, the same four plus nine additional opossums were anesthetized with urethane (125 mg/100 g initial dose, i.p.). Part of the cortex

in one hemisphere was exposed, covered with silicone fluid, and photographed. Recordings were made with low-impedance (1–1.3 M Ω) tungsten microelectrodes, with conventional amplification and monitoring. Electrode penetrations were made perpendicular to the cortical surface, and the location of each penetration was marked on an enlarged photograph of the cortical surface. Recordings were from single neurons and clusters of neurons in the middle layers of the cortex, typically 400–600 μ m from the surface.

Receptive fields for neurons responsive to tactile stimuli were determined by lightly touching the skin with fine probes and displacing hairs and vibrissae. Receptive fields were defined as the region of the body surface where light, near-threshold stimuli produced a clear response. Responses to taps, pressure, and joint movement were also noted. Responsiveness to auditory stimuli was evaluated with clicks produced by tapping metal rods together. Visual receptive fields were determined after the contralateral eye was fixed to a ring to prevent movement and centered in a translucent plastic hemisphere (60 cm in diameter). The optic disc was projected onto the hemisphere as a reference, and projected bars and spots of light were used as stimuli (for more details, see Kaas and Krubitzer, 1992).

Microstimulation of the sensorimotor cortex was carried out under anesthetic conditions identical to those in recording experiments. Because animals under deep anesthesia require higher currents to evoke movements, the level of anesthesia was monitored closely by eyeblink and withdrawal reflexes. Because movement responses in the present experiments did not depend on anesthetic administration schedule (i.e., movements evoked at low thresholds did not correspond to lighter anesthesia), we do not believe high thresholds were due to anesthetic level. Microstimulation was achieved with the same electrodes used for recordings, with the electrode tip at cortical depths of 800–1,000 μ m. Stimulations consisted of 60-ms trains of cathodal current pulses, with a pulse duration of 0.2 ms and a pulse frequency of 300 Hz. Stimulation levels started at 20–30 μ A. If movement was observed, current was decreased progressively until movement ceased. Higher stimulation levels of 40–90 μ A were sometimes used to see if high current levels were effective (for more details, see Stepniewska et al., 1993).

Tracer injections and histology

Injections of 0.02–0.05 μ l of 2% wheatgerm agglutinin-horseradish peroxidase (WGA-HRP) were placed in S1 or in the primary visual area (V1) of three opossums. In a fourth animal, an injection of fluororuby (tetramethyl rhodamine) was placed in S1, and an injection of diamidino yellow was placed rostral to S1 in the presumed somatosensory rostral area (SR; Fig. 1). In cases with S1 injections, sites for injections were first identified by microelectrode recording (see above). In all cases, injections were made under aseptic conditions in anesthetized animals under conditions similar to those used for short-term recording (see above). The animals were killed, typically after a second recording session, 2–3 days later for WGA-HRP injections, and 5 days later after fluorochrome injections.

Animals were given a lethal dose of barbiturate and, when areflexive, perfused transcardially with saline followed by 2–3% paraformaldehyde in phosphate buffer, followed by 2–3% paraformaldehyde and 10% sucrose in phosphate buffer. Brains were removed, and the cortex was separated from the brainstem, unfolded, and flattened.

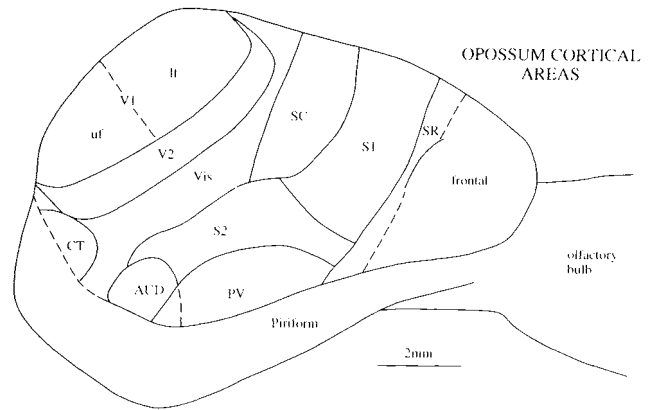


Fig. 1. Proposed subdivisions of the sensory cortex in opossums. Areas are shown on a dorsolateral view of the right hemisphere. Somatosensory areas include primary somatosensory cortex, S1; the second somatosensory area, S2; the parietal ventral somatosensory area, PV; the somatosensory caudal area, SC; and the somatosensory rostral area, SR. The caudal pole is occupied by primary visual cortex, V1, with caudal and rostral portions devoted to the upper visual field, Vif, and lower visual field, Visc. The presumptive second visual area, V2, borders V1 laterally, followed by a band of cortex with visual connections, Visc. More laterally, the cortex responds to auditory stimuli (AUD.) and partially overlaps PV, as indicated with a dotted line. A caudotemporal region, CT, of unknown significance, is distinguished by denser myelination. The frontal and piriform cortices and the olfactory bulb are also indicated.

flattened cortex was stored overnight under a weighted glass slide in 30% sucrose in phosphate buffer. The brainstem was stored separately. The cortex was then sectioned parallel to the surface at 40 μ m with a freezing sliding microtome. Typically, every third section was processed for cytochrome oxidase (Wong-Riley, 1979), myelin (Gallay, 1979), or HRP (tetramethylbenzidine [TMB] reaction; Mesulam, 1978). In some cases, a series of sections were stained with cresyl violet. In the case with the fluorochrome injections, a series of sections were mounted without further processing. The thalamus was cut in the coronal plane at 50 μ m, and series of sections were processed for Nissl, cytochrome oxidase, or HRP.

Illustrated results include drawings of representative sections brain made at low magnification by using a drawing tube attached to a microscope. Such drawings included electrolytic lesions marking recording sites, electrode tracts, injection sites, and major blood vessels. Label from the injections and architectonic borders were aligned across brain sections and transferred to a composite drawing. Reference lesions allowed the pattern of the electrode penetrations to be placed on the composite drawings.

RESULTS

We used microelectrode mapping and stimulation methods in conjunction with the study of cortical connections and architecture to determine subdivisions of the sensorimotor cortex in Virginia opossums. A large middle region of neocortex contained three representations of the body surface: the primary somatosensory area (S1), a second somatosensory area (S2), a parietal ventral area (PV), and two additional fields, here termed the caudal and rostral somatosensory fields (SC and SR), which have somatosensory connections. The locations of these fields are indicated

on a dorsolateral view of the brain in Figure 1. Primary and secondary visual fields, V1 and V2, and a region responsive to auditory stimuli were also identified.

Microelectrode mapping evidence for three somatotopic areas: S1, S2, and PV

Microelectrode recordings provided evidence for three somatotopically organized areas of cortex in each of 10 opossums. The relative locations of electrode penetrations where neurons were responsive to a light touch on the body surface or to the movement of body hairs are shown for eight of these cases in Figures 2 and 3 (for reference, see Fig. 1). For each case, additional recording sites were placed in adjoining regions of the cortex that were not responsive to cutaneous stimuli. In the figures, area boundaries were estimated by drawing borders midway between two adjacent recording sites determined to be in different areas. Thinner lines separate clusters of recording sites where neurons were responsive to stimuli on different body parts. For locations with few recording sites, lines separating representations of body parts were inferred from patterns observed in other cases, and the outer boundaries of areas were determined, where possible, by architectonic distinctions (see below). For example, in case 92-51, the most medial cluster of seven recording sites (indicated with dots in Fig. 2) encountered neurons responsive to light touch on parts of the hindlimb, whereas neurons at progressively more lateral sites were activated by stimulating the trunk, forelimb, chin, mystacial vibrissae, rhinarium, or lips. Two additional representations were found in the cortex caudolateral and lateral to these sites, thereby providing evidence for three representations of the body surface.

The primary somatosensory area, S1. One representation had the expected location and somatotopic order of S1 (Kaas, 1983). As with other mammals, S1 of opossums represented the body from hindlimb to nose in a medial-to-lateral sequence. Figure 4 shows receptive fields for a selected row of recording sites across the mediolateral extent of S1 in case 92-51 (other recording sites are marked in Fig. 2) in which the characteristic somatotopic progression from hindlimb to nose was revealed. Receptive fields were located on the foot and hindleg for the most medial recording sites, followed by the trunk and forearm, the dorsal and glabrous forepaw, lateral face, mystacial vibrissae, and rhinarium most laterally. Other organizational features of lateral S1 are documented in Figure 5. The curving mediolateral progression of recording sites relates to a sequence of receptive fields starting on the dorsal forearm and moving to the nailbed of a single forepaw digit, the glabrous tips of two adjoining digits, caudal and then rostral mystacial vibrissae, and, finally, the rhinarium and lips. In Figure 5, receptive fields for a rostralward progression of penetration sites 6–9 indicated that the midline of the rhinarium was represented caudal to more lateral parts of the rhinarium and face. Likewise, a caudorostral row of sites, 10–13, indicated that dorsal vibrissae near the dorsal midline of the face were represented caudal to ventral vibrissae and that rostral S1 represented the ventral midline of the face on the lower jaw. Additional receptive fields for neurons in lateral S1 are shown in Figure 6. Again, it was apparent that caudolateral S1 was devoted to mystacial vibrissae and the rhinarium. Overall, the revealed somatotopic features of S1 closely approximated those previously indicated for most of S1 of opossums and, indeed, for S1 of

other species. Thus, we were able to locate and identify S1 reliably and to approximate external boundaries and internal somatotopic subdivisions. In general, somatotopic features of S1 in the present study closely correspond to those described previously for S1 of opossum (Pubols et al., 1976), with the exception that we included a caudolateral second representation of parts of the head in S2 rather than S1 (see below and Discussion).

The second somatosensory area, S2. We identified a second systematic representation of the body surface in the cortex caudolateral to S1 as S2. Much of this cortex was previously included in S1 as a second representation of the rostral face within S1 (Pubols et al., 1976). The nature of this double representation was apparent from the receptive field locations for recording sites in lateral S1 and medial S2 (Fig. 6). Receptive fields for progressively more lateral sites in S1 started on the lower face (site 1), moved to the mystacial vibrissae (site 2), and ended on the rhinarium (sites 3–5). The neurons at the next site in the progression had much larger receptive fields covering much of the rhinarium. The larger receptive fields included skin covered by the smaller receptive fields for sites 3–5. Subsequent receptive fields for the lateralward (and caudalward) progression of recording sites included the rhinarium, vibrissae, and more caudal face. Further evidence of the S1–S2 border was apparent from results illustrated for a second row of more rostral recording sites, 11–17, in Figure 6. Starting with site 11 in S1, with a receptive field on the chin vibrissae, sites 12–13 again demonstrated a progression of receptive fields onto the rhinarium. The receptive field for site 14 was larger than those for sites 12 and 13, and sites 14–17 corresponded to a progression of receptive fields from the rhinarium to the lateral face just under the eye, thus revealing a re-representation of this part of the face in the cortex caudolateral to S1. A reversal of the somatotopic order and an enlargement of receptive fields were two characteristic features for neurons in rows of recording sites crossing from S1 to S2 in several species (e.g., Krubitzer & Kaas, 1989). Thus, in case 93-06, we placed the border of S1 with S2 between sites 5 and 6. Reversal of the somatotopic order occurred consistently between S1 and S2 in opossums and thus was used as a strict criterion in determining the border between these two areas. Enlargement of receptive fields almost always coincided with reversals in somatotopic order and was used as a second criterion. However, occasionally, receptive field sizes were similar in S1 and S2, especially on the vibrissae, so when a change in receptive field size did not coincide with a somatotopic reversal, only the latter was used to delineate the border.

Other somatotopic features of S2 are shown in Figures 7–9. The results indicate that the rhinarium, facial vibrissae, and caudal face occupied much of S2 in a rostrocaudal representational sequence. A small caudoventral portion of S2 represented the limbs and trunk. Thus, a dorsoventral row of sites in caudal S2 (sites 11–14; Fig. 7B) corresponded to receptive fields on the upper head, lateral face, trunk, and hindlimb. Receptive fields for some of these sites were extremely large, with neurons at site 13 (Fig. 7B) being activated from hair movement and a light touch over most of the contralateral body. In contrast, receptive fields for some sites (e.g., site 3 in Fig. 7B) in the mystacial vibrissae region of S2 had receptive fields involving only a few vibrissae and thus were not notably larger than comparable receptive fields for S1. Receptive fields on the distal forepaw

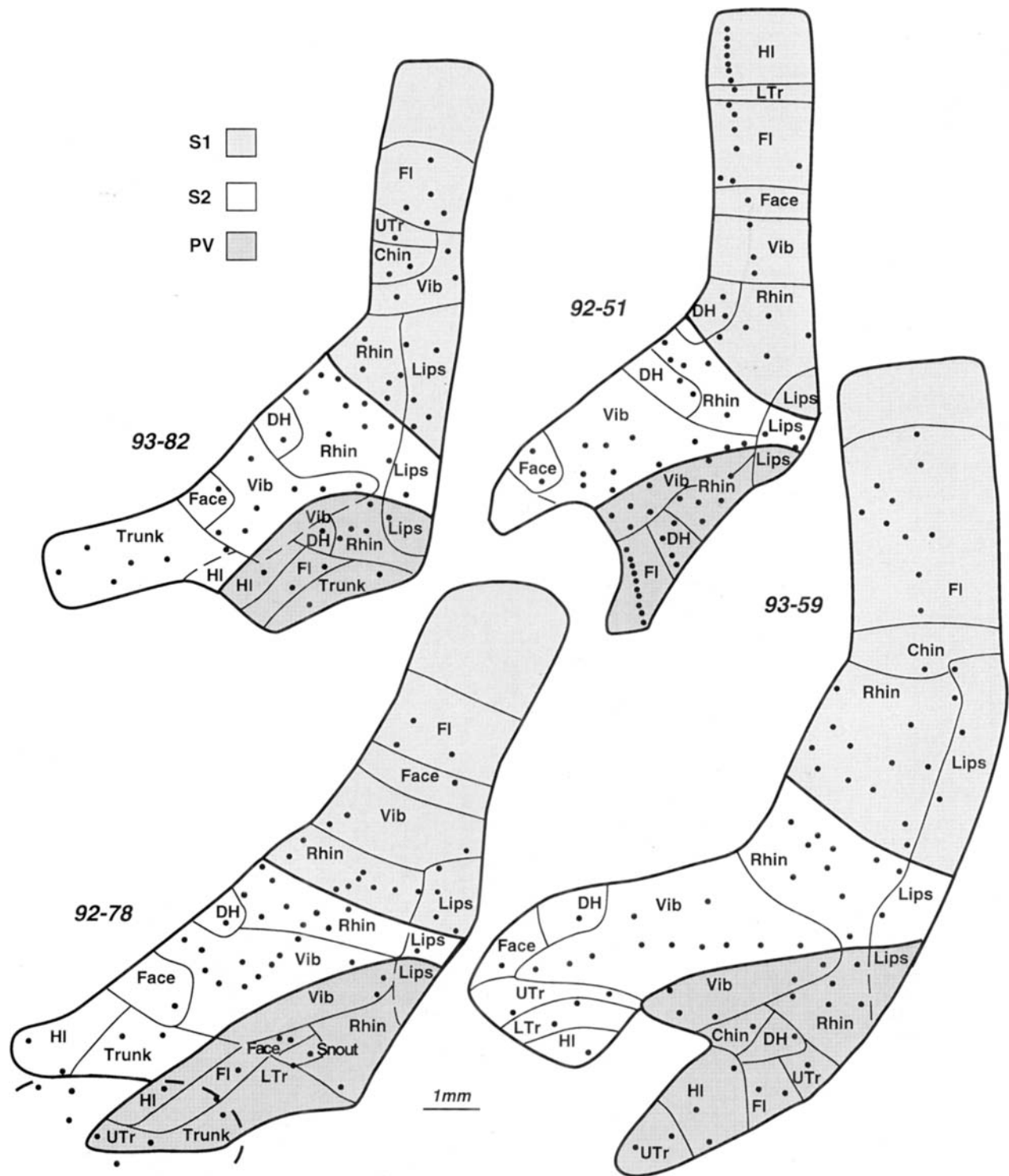


Fig. 2. The somatotopic organizations of S1, S2, and PV in four opossums. The outer boundaries of S1, S2, and PV (see Fig. 1) were estimated from somatotopic patterns and cortical architecture for each numbered case. Within each field, lines separate recording sites (dots), with neurons activated by a light touch on a designated body part. The location of these lines are estimates that are most accurate in regions with closely spaced recording sites. Borders between areas are best estimates based on changes in somatotopy, receptive field characteristics, and architecture. The results indicate that each area represents

most or all of the contralateral body surface in a topographic manner. Results from these and other cases were sometimes obtained from the left hemisphere, but they are all illustrated as if from the right hemisphere for ease of comparison. Some sites in S2 and PV and numerous sites caudal to these fields responded to auditory stimuli. HI, hindlimb; LTr, lower trunk; FI, forelimb; UTr, upper trunk; Vib, vibrissae; Rhin, rhinarium; DH, dorsal head and snout.

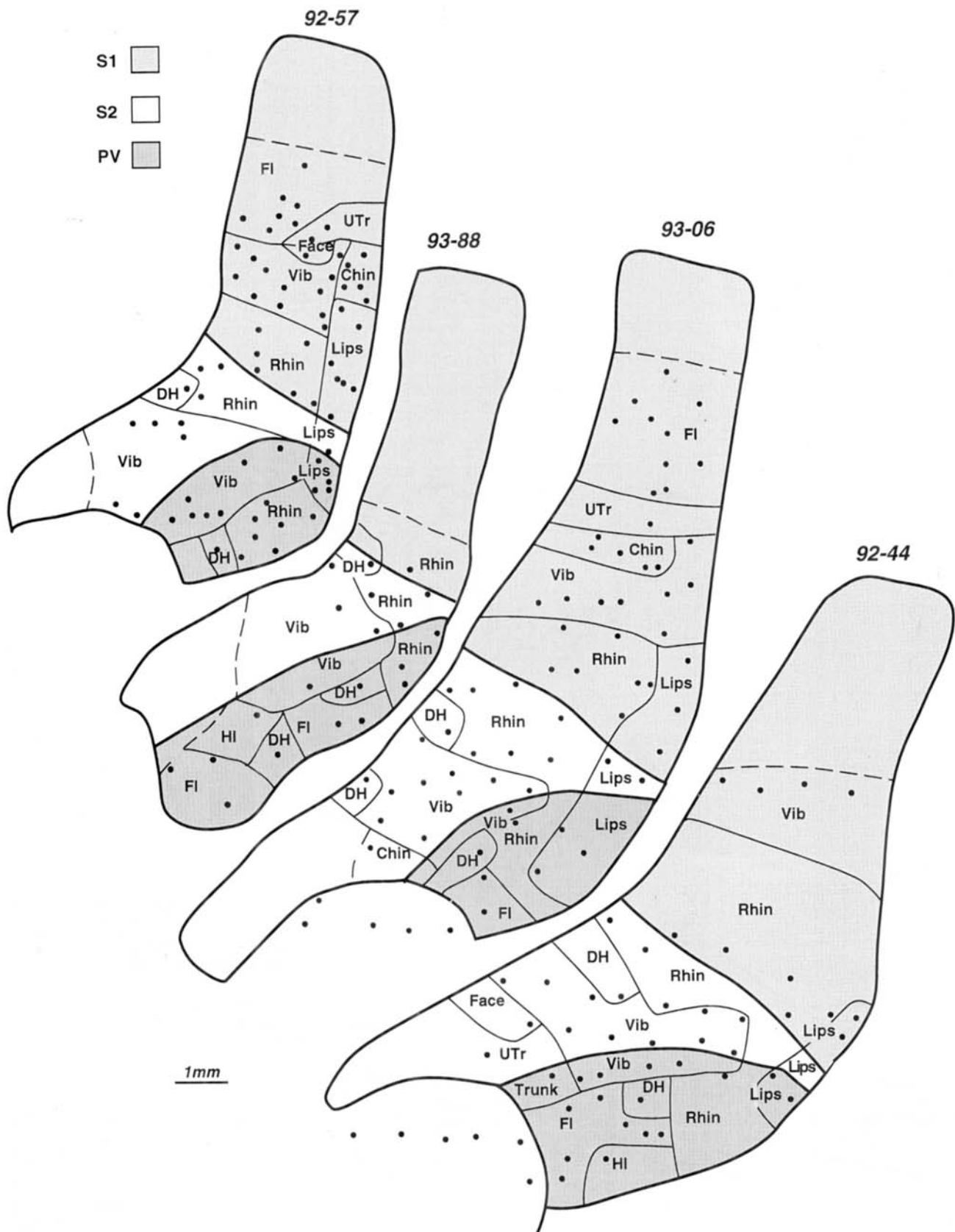


Fig. 3. Somatotopy of S1, S2, and PV in four additional opossums. Recording sites caudal to PV responded to auditory stimuli. Conventions as in Figure 2.

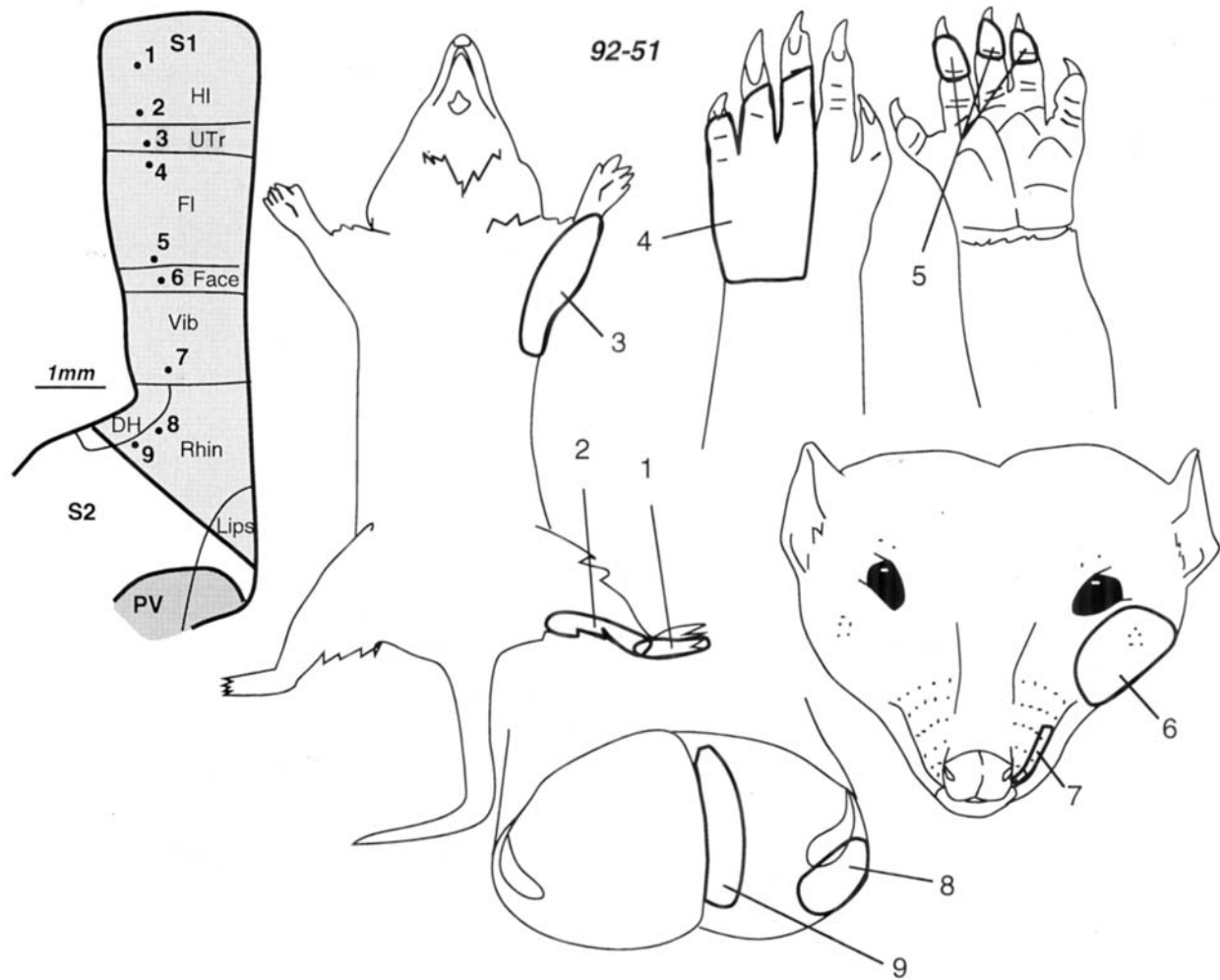


Fig. 4. The mediolateral somatotopic organization of S1. Receptive fields for a mediolateral row of recording sites demonstrate a progression of location from the hindlimb to the tip of the nose. Receptive fields

are numbered according to recording site (dots) indicated on a surface view of S1 and the adjoining parts of S2 and PV. Compare with Figures 1 and 2.

were not detected in these experiments. The expected location of the forepaw would be in the caudal tip of S2, a region we sampled rather poorly.

The parietal ventral somatosensory area, PV. The body surface was represented a third time in the cortex located just ventral to S2. Because this representation had the relative location and overall somatotopic organization of a previously described parietal ventral somatosensory area, we termed this area PV (see Krubitzer et al., 1986; Discussion). The representation in PV paralleled that of S2 in rostrocaudal somatotopic organization, so that it was sometimes difficult to assign recording sites along the common border to one or the other area with confidence. Nevertheless, we were able to record from enough sites in the two areas in several animals (Figs. 2, 3) that two parallel representations were evident, and the evidence for PV as an area separate from S2 was compelling.

Receptive fields for some of the recording sites of case 93-59 are shown in Figure 8. Receptive fields in both S2 and PV were often similar in size and quite large, but crude reversals in somatotopy could be used in combination with architecture to estimate borders. Receptive fields for sites

1-4 revealed a reversal in somatotopic organization across the S2-PV border, with an increase of receptive field size for sites in PV. Thus, the rhinarium and mystacial vibrissae were represented dorsally in S2 and again ventrally in PV. Similarly, receptive fields for sites 5-9 progressed from the upper face along the bridge of the snout to facial vibrissae in S2 and from the facial vibrissae to the upper face along the bridge of the snout in PV, with a reversal in the somatotopic progression occurring between sites 6 and 7. However, a notable increase in receptive field size for neurons at site 9 was not apparent, indicating that neurons in PV do not consistently have larger receptive fields than neurons in S2. Finally, receptive fields for a third dorsoventral row of recording sites (11-16) showed that the limbs and trunk were represented dorsally in S2 and, again, ventrally in PV. The cortex between sites 13 and 14 was unresponsive to somatosensory stimuli but was responsive to auditory stimuli. Thus, somatosensory areas border the auditory cortex.

Receptive fields are shown for additional recording sites in PV in Figures 9 and 10. Receptive fields illustrated in Figure 9 clearly indicated that the trunk was represented

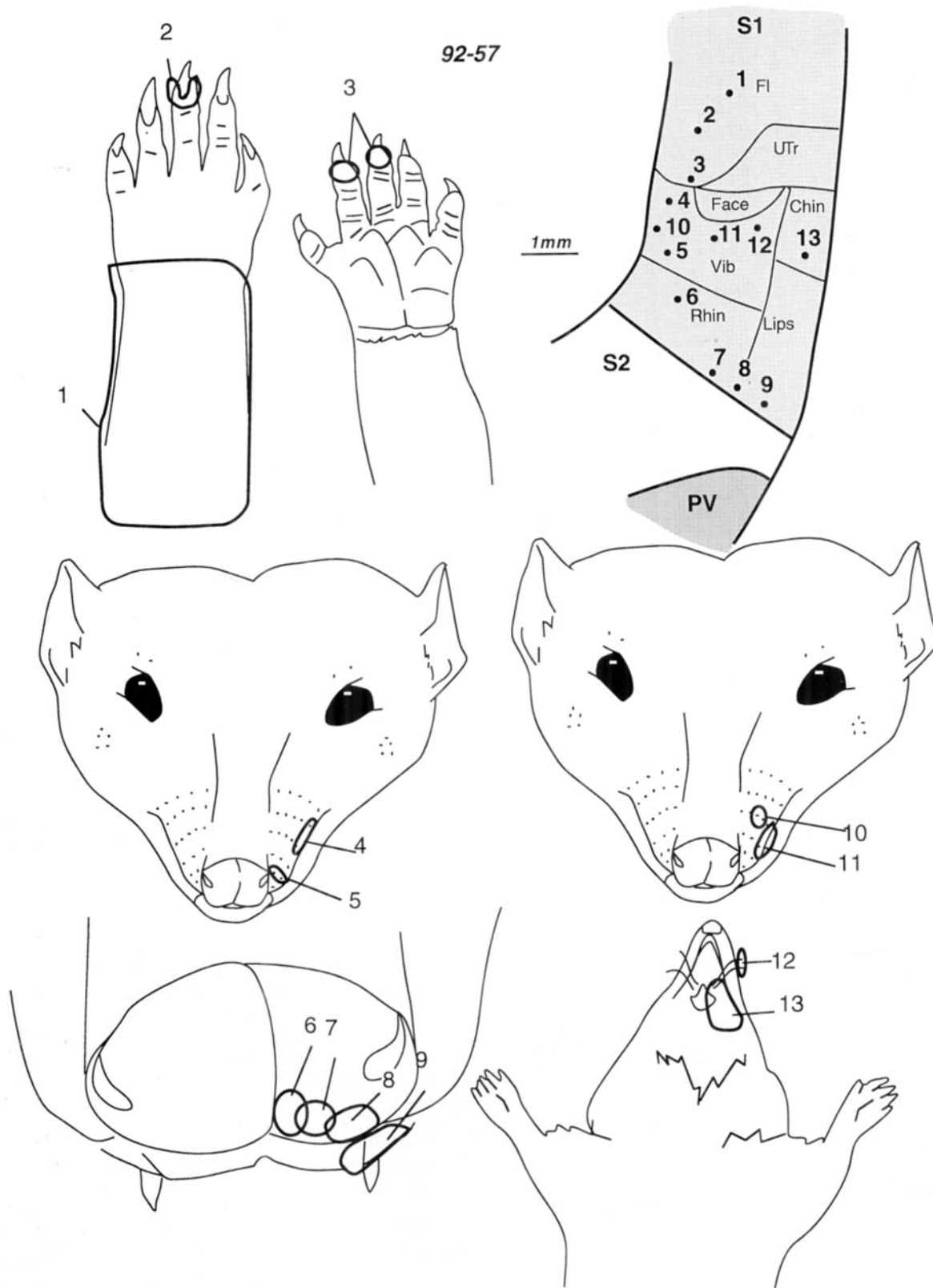


Fig. 5. The somatotopic organization of lateral S1. Receptive fields are shown for neurons in a mediolateral row of electrode penetrations, 1-9, that curves rostrally along the border of S1. A second row of

penetrations, 10-13, crosses the width of S1 and illustrates that rostral S1 relates to the ventral body. Other conventions as in Figure 4. Compare with Figures 1 and 3.

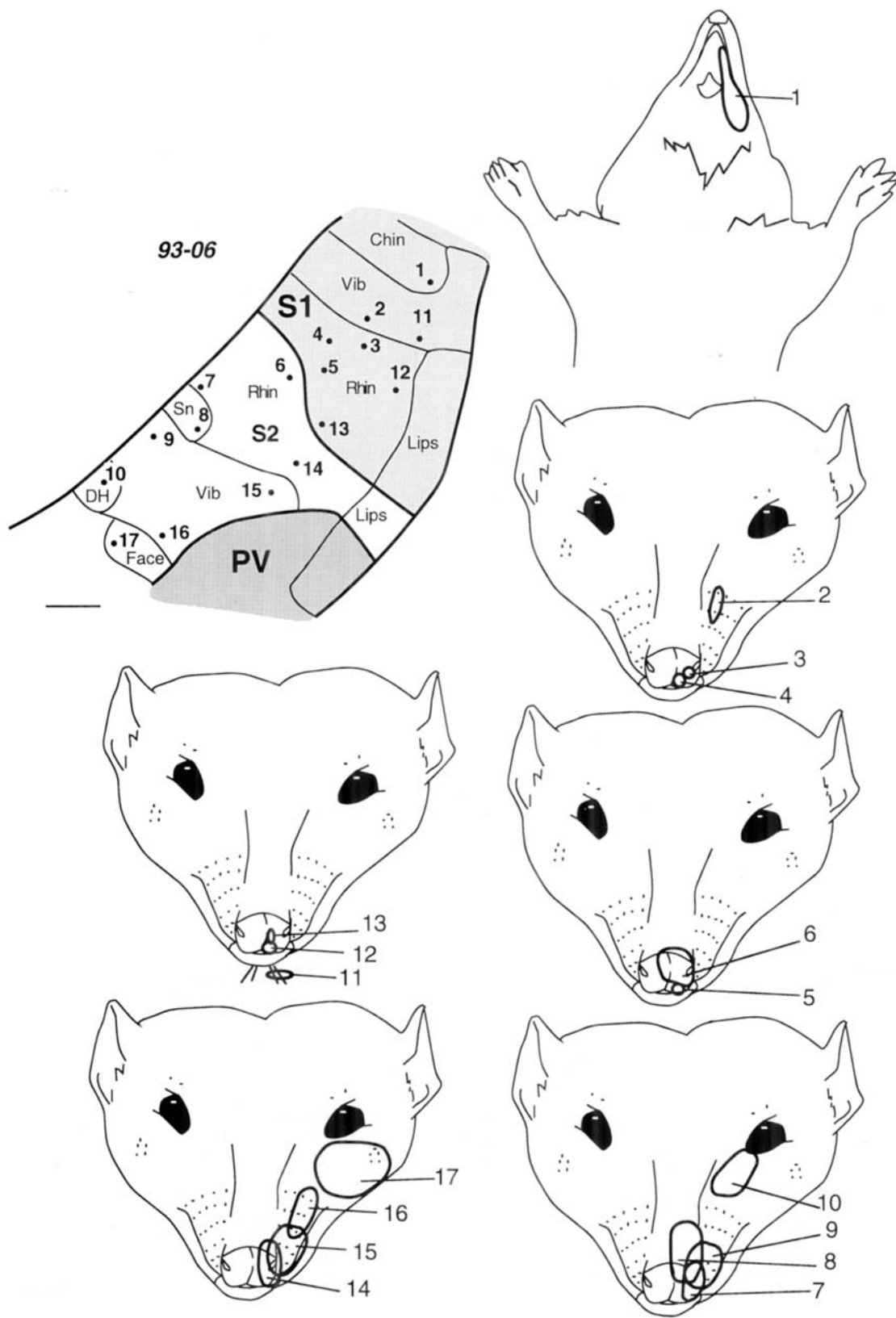


Fig. 6. Receptive fields for neurons in rows of electrode penetrations crossing from S1 to S2. Somatotopic progressions of receptive fields reverse and receptive fields become larger for sites in S2. See rows 1–10 and 11–17. Compare with Figures 1 and 3. Conventions as in Figure 4. Scale bar = 1 mm.

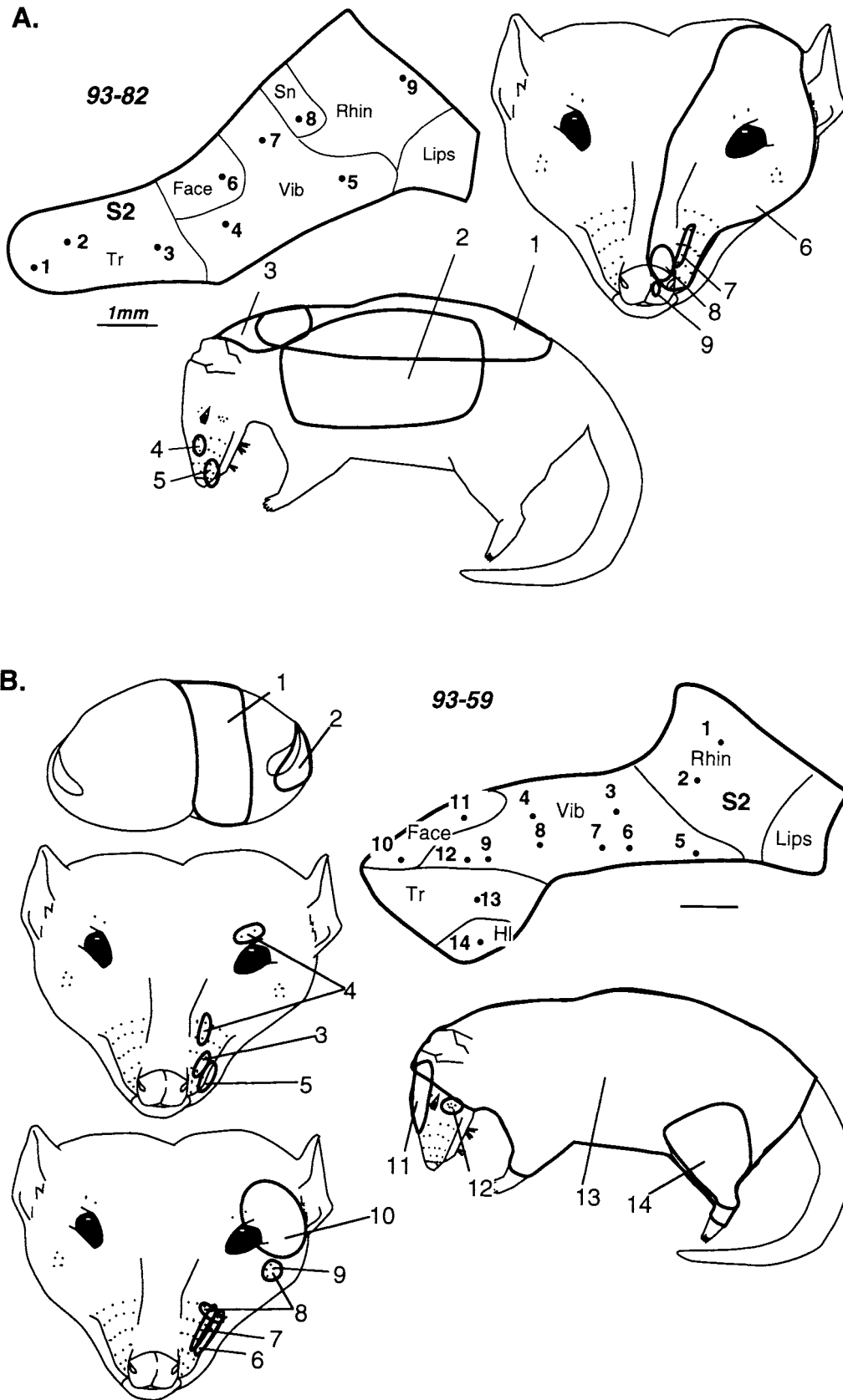


Fig. 7. Receptive fields for neurons in electrode penetrations in S2 in two cases, 93-82 (**A**) and 93-59 (**B**). Conventions as in previous figures. Scale bars = 1 mm.

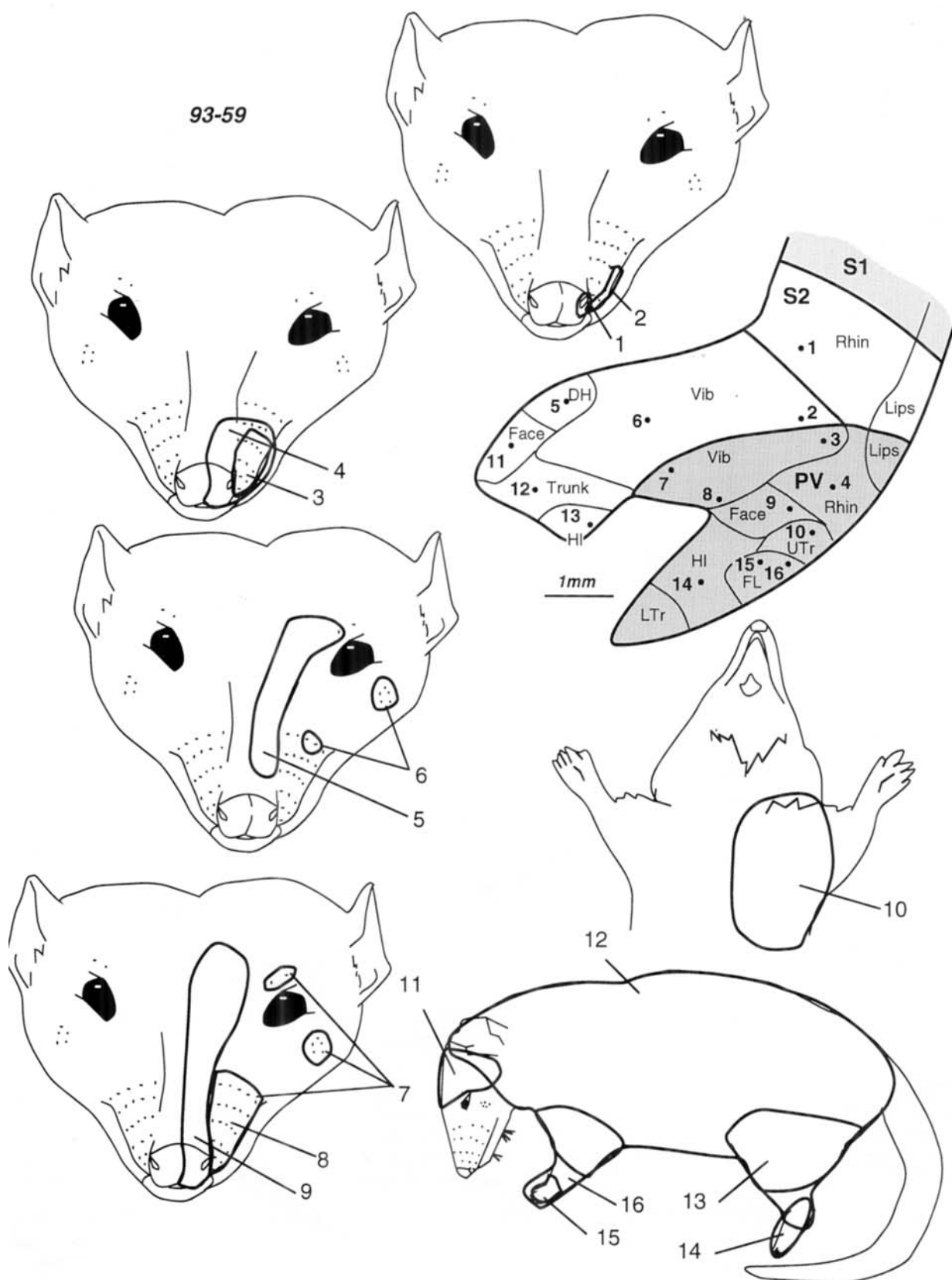


Fig. 8. Receptive fields for neurons in electrode penetrations in three rows crossing from S2 to PV (1-4, 5-10, 11-14). Conventions as in previous figures.

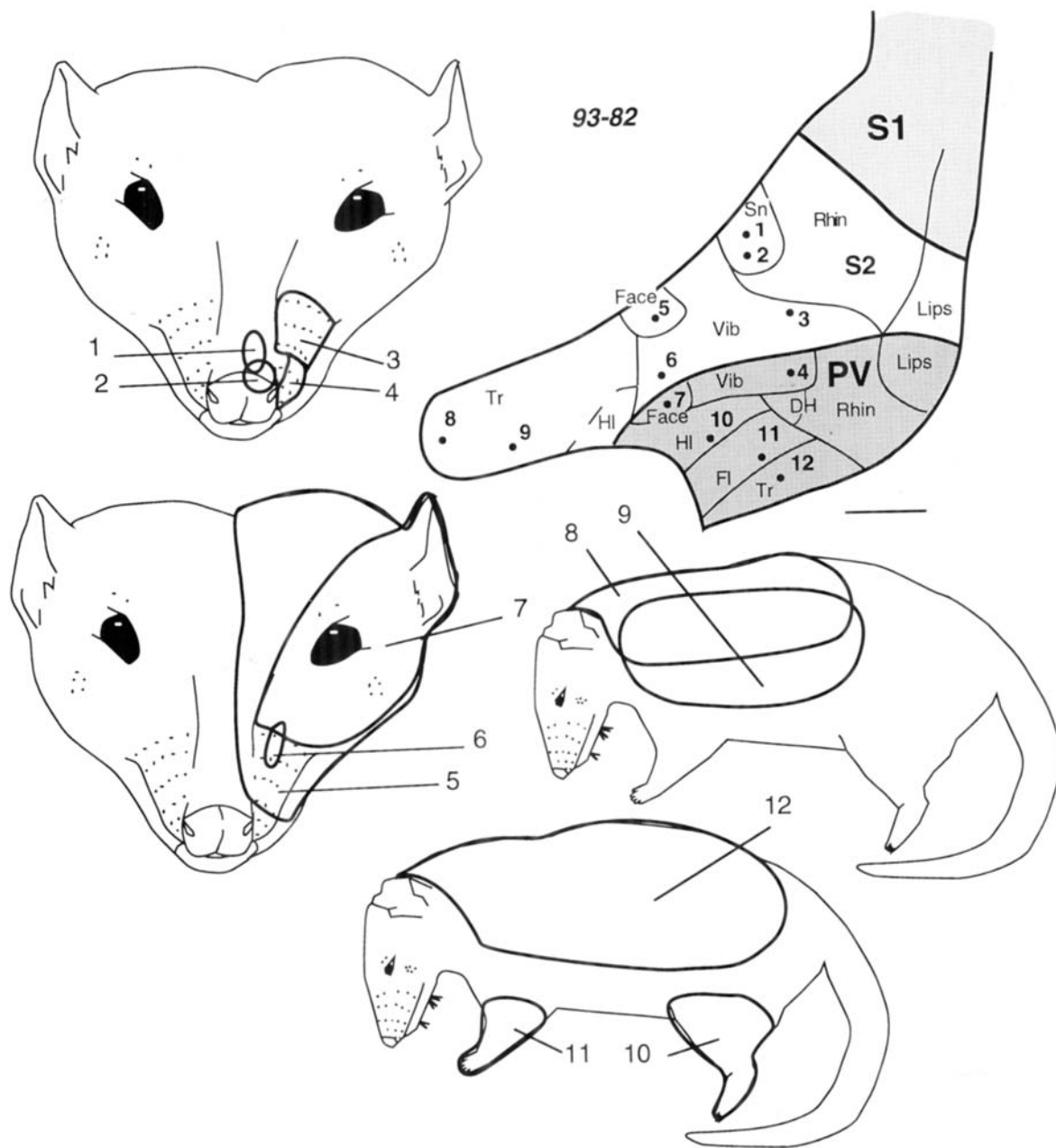


Fig. 9. Receptive fields for neurons in electrode penetrations in three rows crossing from S2 to PV. Conventions as in previous figures. Scale bar = 1 mm.

dorsally in S2 (sites 8 and 9) and again at some distance ventrally in PV (Fig. 9, site 12). Within PV, receptive fields on the trunk were quite large (Fig. 9, site 12; Fig. 10A, site 1), but somatopy was evident. Receptive fields on the forepaw varied from being confined to a few digits, covering the glabrous palm, to including hairs extending from the dorsal forepaw to the elbow (Fig. 10A, sites 5 and 7). Neurons at site 6 (Fig. 10A) were activated from both the forefoot and the hindfoot.

Cortex caudal and rostral to S1. Although the cortex caudal and rostral to S1 was not mapped in great detail, a small number of recording sites in these regions responded to somatosensory stimuli. Out of 60 recording sites within 2

mm caudal to S1 (SC), only five sites responded to cutaneous stimuli and eight sites responded to taps. Forty-six sites in the cortex between S1 and the orbital sulcus (SR) were examined for responsiveness to somatosensory stimuli. Two of these sites responded to cutaneous stimuli, and seven responded to taps.

Cutaneous responses were found caudal to S1 in three animals, and noncutaneous responses were found in five animals. In case 92-44, one site out of three examined had a large receptive field that included the snout, lateral face, and dorsal head. Responses habituated within 10 rapid strokes to the skin. Two (of 13) sites responded very weakly to cutaneous stimuli on the lateral face in case 92-57. In a

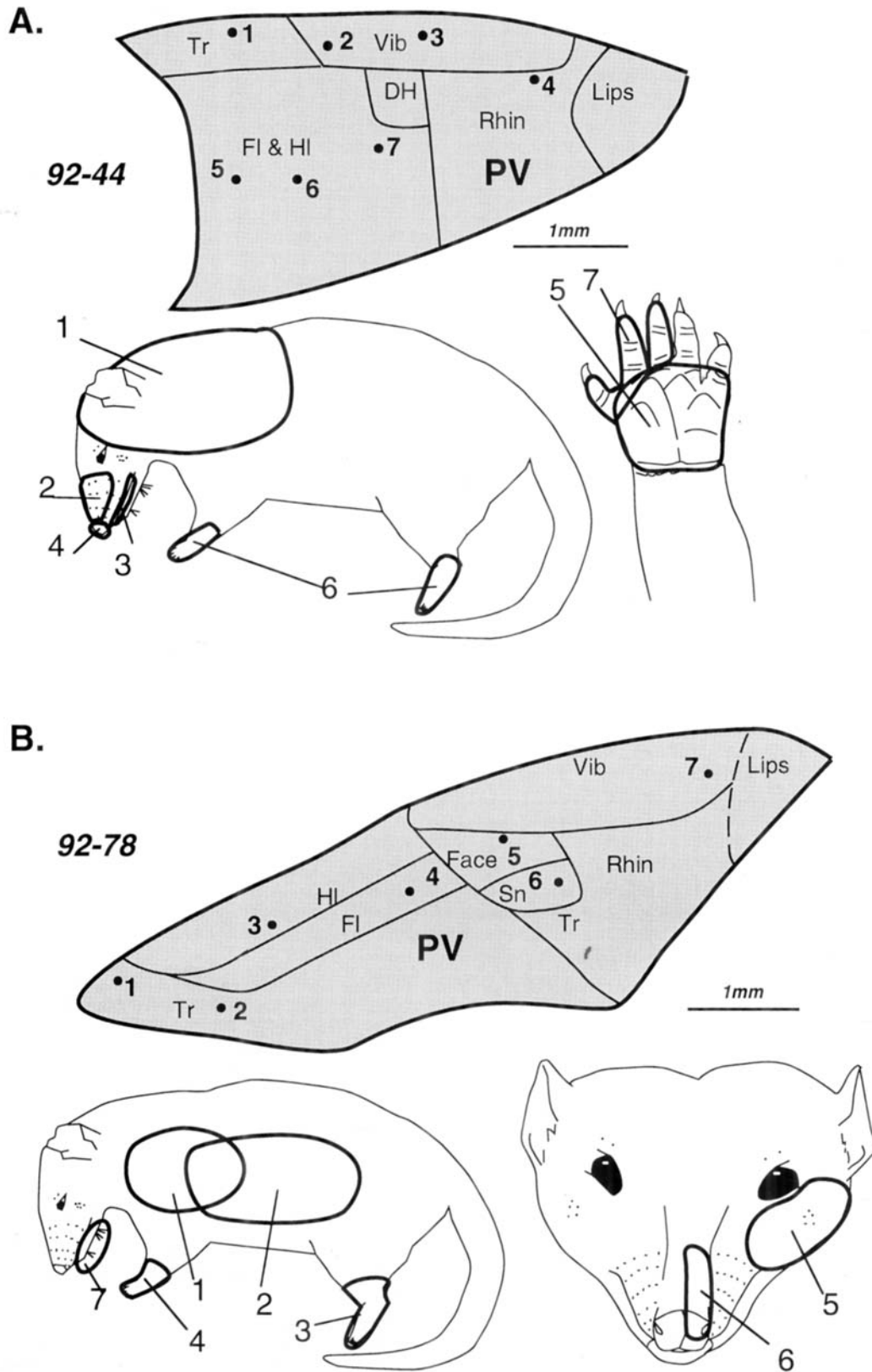


Fig. 10. Receptive fields for neurons in electrode penetrations in PV in cases 92-44 (**A**) and 92-78 (**B**). Conventions as in previous figures.

third case (93-82), two (of 17) sites responded very weakly but consistently to stimulation of the dorsal snout. Responses to hard taps to the hand were found caudal to S1 in case 92-51 in one of four sites examined. One of 13 sites in case 92-57, one of three sites in case 93-59, and one of nine sites in case 93-88 responded to taps to the nose. Responses to taps of the lips and/or chin were found caudal to S1 at one of three sites in 93-59, one of 17 sites in case 93-82, and one of nine sites in 93-88. One of 17 sites examined caudal to S1 in case 93-82 responded to taps to the dorsal snout. In eight animals, 46 of 60 recording sites caudal to S1 did not respond to somatosensory stimuli.

In the cortex rostral to S1 and the S2-PV junction, somatosensory responses were found in three of eight animals. In case 92-44, one of seven sites responded weakly to cutaneous stimulation on the lower trunk and hip. In case 93-88, one of 14 sites, located at the S1-SR border, responded to stimulation of the palate and tongue. Because responses at the rostral border of S1 are typically to stimulation of oral structures and because the response is cutaneous, this site could be located within an S1 border that protrudes rostrally. Noncutaneous responses were found in two animals. One of seven sites examined rostral to S1 in case 92-57 responded to taps to the lower lip. In case 93-88, six of 14 sites responded to noncutaneous stimuli. One site responded weakly to taps to the lower lip, one to taps to the chin, two responded weakly to taps of the hand, and one to taps of the distal digits. Only one site (case 93-88) in eight animals responded to movement, particularly movement of the jaw.

Auditory and somatosensory overlap. For many recording sites, we tested neurons for responsiveness to auditory and somatosensory stimuli. Sites within an oval of cortex just caudal to PV and partially overlapping caudal PV consistently responded to sounds produced by tapping two objects together. Occasionally, a site in caudal S2 also responded to auditory stimuli. Although we did not stimulate with pure tones or attempt to determine the tonotopic organization of the region responsive to auditory stimuli, the auditory region could be divided into a solely auditory portion, possibly A1, and another portion overlapping with PV and sometimes S2. Somatosensory fields from the overlap region are shown in Figure 11. Neurons at recording sites (1-5), indicated with asterisks, were responsive to auditory and cutaneous stimuli. Similar to previous descriptions of PV, the somatosensory receptive fields for these neurons were large and located on the limbs and trunk (Krubitzer et al., 1986; see Discussion).

Movements produced by microstimulation at sites within and around S1. The cortex was electrically stimulated with microelectrodes at 181 sites in three opossums. Levels of current (5-20 μ A) that readily evoke body movements from motor cortex in monkeys and other mammals (e.g., Stepniewska et al., 1993) generally failed to evoke movements from the somatosensory cortex or the more rostral cortex in these opossums, producing movements in only 54 of the 181 sites stimulated (Fig. 12). Movements were not evoked at most sites, even with much higher current levels of up to 100 μ A. Nevertheless, in case 93-86 (Fig. 12A), jaw and tongue movements were produced at eight sites in lateral S1, five sites in SR, and two sites rostral to SR and lateral to the orbital sulcus. Elbow and shoulder movements were evoked at two sites in lateral S1, one site in SR, and one site rostral to the orbital sulcus. Sites in S1 were also tested for responsiveness to tactile stimuli, and move-

ment fields and receptive fields were roughly in register. Likewise, wrist, tongue, and jaw movements were evoked from nine sites in lateral S1, two sites in SC, five sites in SR, and two sites rostral to the orbital sulcus (eye movements) in case 93-88 (Fig. 12B). In a third case, 93-83, movements of the elbow and shoulder were evoked at two sites in S1, and movements of the tongue were evoked at seven sites in S1. Two sites just rostral to S1 produced movements of the elbow and shoulder, and one site evoked movement of the tongue. Six sites just rostral to the S1-S2 junction evoked movements of the tongue. The results implicate the cortex rostral to S1 in motor function, but they do not provide clear evidence for a specialized motor area in the usual location of M1 in cortex (rostral to S1) or in the cortex that is coextensive with S1, as proposed by Lende (1963c).

Architecture of somatosensory areas

Electrophysiological results were related to the cortical architecture in brain sections cut parallel to the surfaces of the manually flattened, experimental cerebral hemispheres. This procedure allowed a direct comparison of architectonic and physiological patterns, although laminar differences in architecture were less apparent. In two cases, we stained successive series of sections for cell bodies (Nissl), cytochrome oxidase (CO), or myelin.

Nissl preparations usually did not clearly reveal borders of different areas, but some subdivisions of the cortex were nevertheless apparent. Primary visual cortex (V1) was conspicuous in sections cut through the middle layers of the cortex as a darker region resulting from the denser packing of small neurons in layer IV, a feature of the sensory cortex. Other cortical regions, including S1, S2, PV, and auditory cortex, were darker in Nissl-stained sections, but the borders of these subdivisions were not always apparent.

The CO preparations revealed subdivisions more clearly (Fig. 13A). Again, V1 was obvious as a dark oval. In some sections, the region depicted as V2 (Fig. 1) was apparent as a series of CO-light ovals with CO-dark surrounds, as has been described for South American opossums (Martinich et al., 1990). The primary somatosensory area, S2, PV, and auditory cortex constituted a larger CO-dark region, which extended caudally to near the rostral tip of V1. This caudalward extension of the CO-dark region included the cortex we refer to as the caudal somatosensory area, SC (Figs. 1, 13A). However, SC was somewhat less CO-dense than S1, and PV was less CO-dense than S2. The auditory core, possibly A1, tended to be a little darker in CO preparations.

Within lateral S1 and the adjoining part of S2, the density of the CO staining was clearly uneven when examined at higher magnification (Fig. 14). Because these portions of S1 and S2 represent vibrissae and other parts of the face, we assume that the CO-dense ovals and patches correspond to concentrations of receptors on the face related to specific vibrissae and other structural subdivisions, much as the CO pattern reveals the representations of individual mystacial vibrissae and other body regions in S1 of rats (e.g., Dawson and Killackey, 1987) and the rays of the nose in S1 and S2 of star-nosed moles (Catania and Kaas, 1995). However, we have not related individual CO-dense ovals to specific body structures.

Finally, the global pattern of myelin staining (Fig. 13B) was useful in demarcating separate cortical areas, but the density of staining varied appreciably within areas accord-

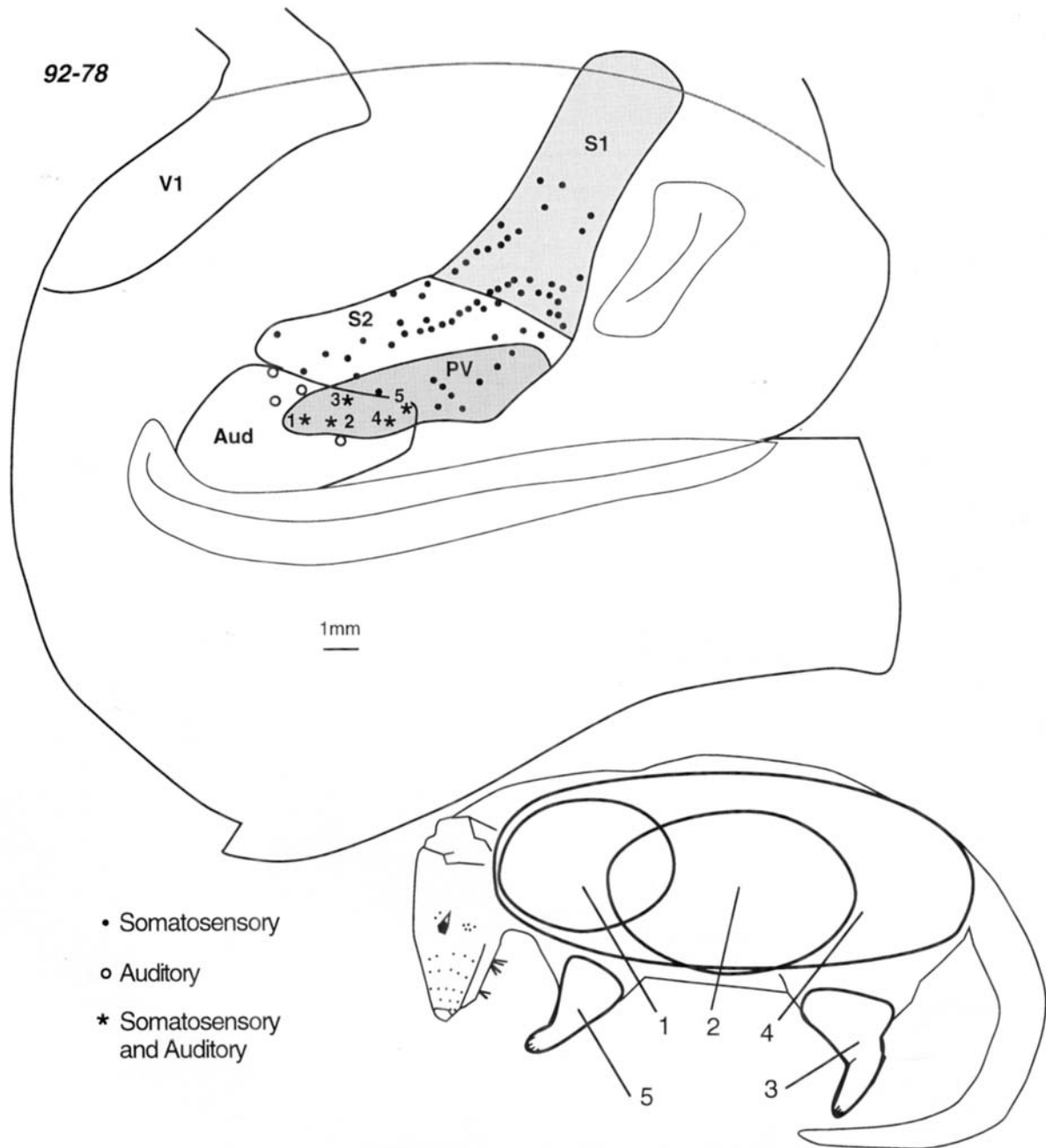


Fig. 11. Somatosensory receptive fields for neurons that also responded to auditory clicks. Most of these bimodal neurons were in PV.

ing to what layers were included in the section. Thus, it was especially important to consider several sections when judging boundaries. In favorable sections, the S1, S2, PV, and auditory regions stood out as densely myelinated. The primary visual cortex was light and indistinguishable from the surrounding cortex in sections through superficial layers and dark through the band of Genarri (Fig. 13B). The retrosplenial area on the medial wall was quite dense, as in other mammals (e.g., Zilles, 1990). Thus, all three preparations distinguished the primary visual cortex and a conglomerate of the somatosensory and the auditory cortex.

Connections of S1

Connections of parts of lateral S1 were revealed by single injections of WGA-HRP in three opossums, and two injections of fluororuby and diamidino yellow in one opossum. The injections were placed at physiologically identified sites and were largely or completely confined to S1. Because the cortex was cut parallel to the surface, the major features of the labeled connection pattern could be appreciated from the distribution of label within single favorable sections. In Figure 15, for example, a darkfield photomicrograph of a brain section through the middle layers of the cortex shows

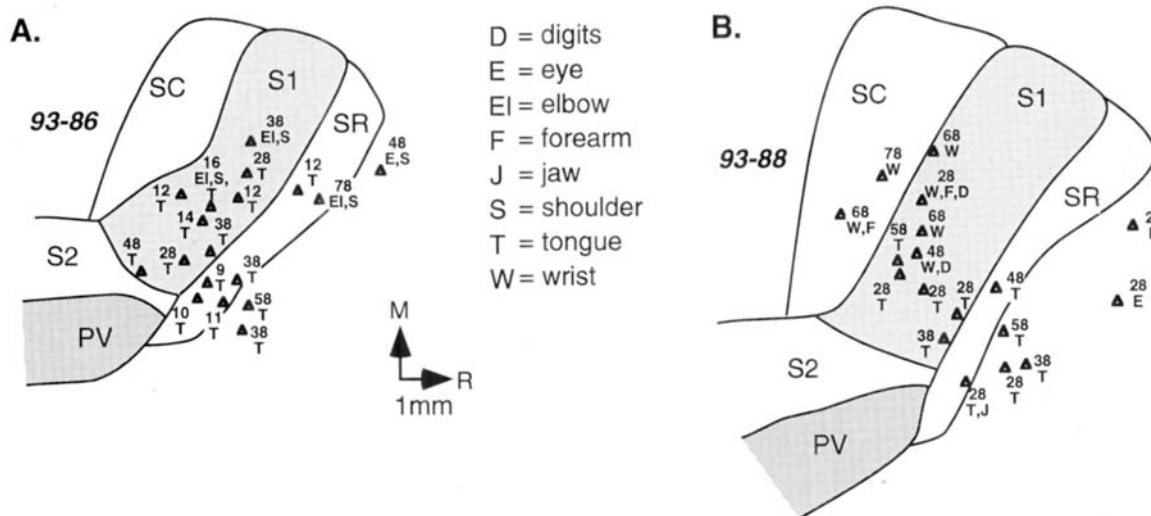


Fig. 12. Penetration sites at which movements were evoked by microstimulation in two cases, 93-86 (A) and 93-88 (B). Triangles indicate each site at which movement thresholds were below 100 μ A and are superimposed on somatosensory areas determined by physiologi-

cal recordings (indicated with conventions used in previous figures) to demonstrate the relative locations of motor and somatosensory fields. The region of the body from which movement occurred (small letters) and movement thresholds (small numbers) are noted for each site.

a WGA-HRP injection site in S1 and transported label in S2, PV, SC, and SR. In the photomicrograph, the injection site is marked by an obvious puncture in S1 that is surrounded by a dense, dark circle of label that is generally presumed to be the effective uptake zone of the injection (e.g., Mesulam, 1982). Although the injection puncture was placed at a site where microelectrode recordings encountered neurons responsive to a touch to the chin, the uptake zone included cortex representing the facial vibrissae and parts of the forelimb. The dense core of the injection site covered most of the width of S1 and may have marginally involved the rostral border of S1. However, because label was similar to cases in which the injection core did not extend to the rostral border of S1, we do not believe label in SR is due to local connections of SR. Less dense label in fibers and neurons formed a belt of short, intrinsic connections around the injection core within S1. Labeled short-range connections among S1, SC, and SR appeared as rostralward and caudalward protrusions from the injection core. Because of uneven flattening of the cortex in the region of the orbital dimple, the rostralward protrusions appeared to end sharply. However, these bands of label continued farther rostrally in deeper brain sections. Distinctly separate foci of label were also apparent in S2 and PV. However, the patch of label just rostral to PV was comprised of labeled fibers in the white matter just under the cortex. These fibers were revealed because the main section unevenly crossed cortical layers near the rhinal fissure. Thus, the labeled fibers represented those connecting S1 with subcortical structures rather than a zone of cortical connections.

A reconstruction of the labeled zones based on several brain sections revealed additional features of the connection pattern (Fig. 16). Although individual layers were difficult to discern in flattened sections, both fine processes and labeled cells were found in every section, extending from the pial surface to the white matter, reacted for HRP. Except for the labeled fiber bundle near the rostral tip of PV, all regions of label included both fine processes, assumed to be axon terminations, and cell bodies. Thus, local

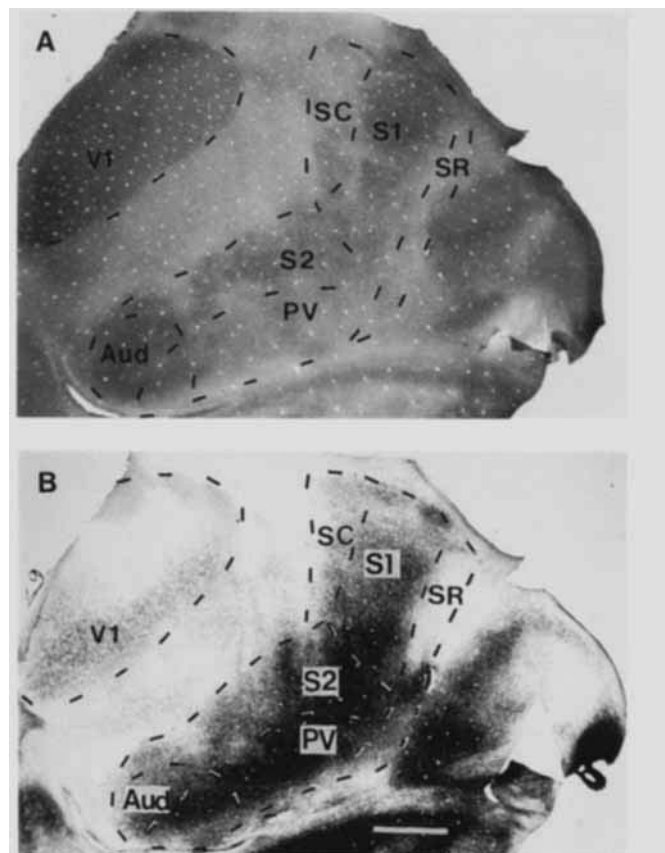


Fig. 13. Architectonic subdivisions of the sensory cortex. A: Somatosensory areas S1, S2, PV, and SC, the auditory cortex, and V1 react more darkly than the surrounding cortex for cytochrome oxidase. SR corresponds to a band of cortex just rostral to S1, which stains lightly for cytochrome oxidase. B: Somatosensory areas S1, S2, and PV, and the auditory cortex are densely myelinated. Somatosensory areas SC and SR and V1 are less densely myelinated. Scale bar = 3 mm.

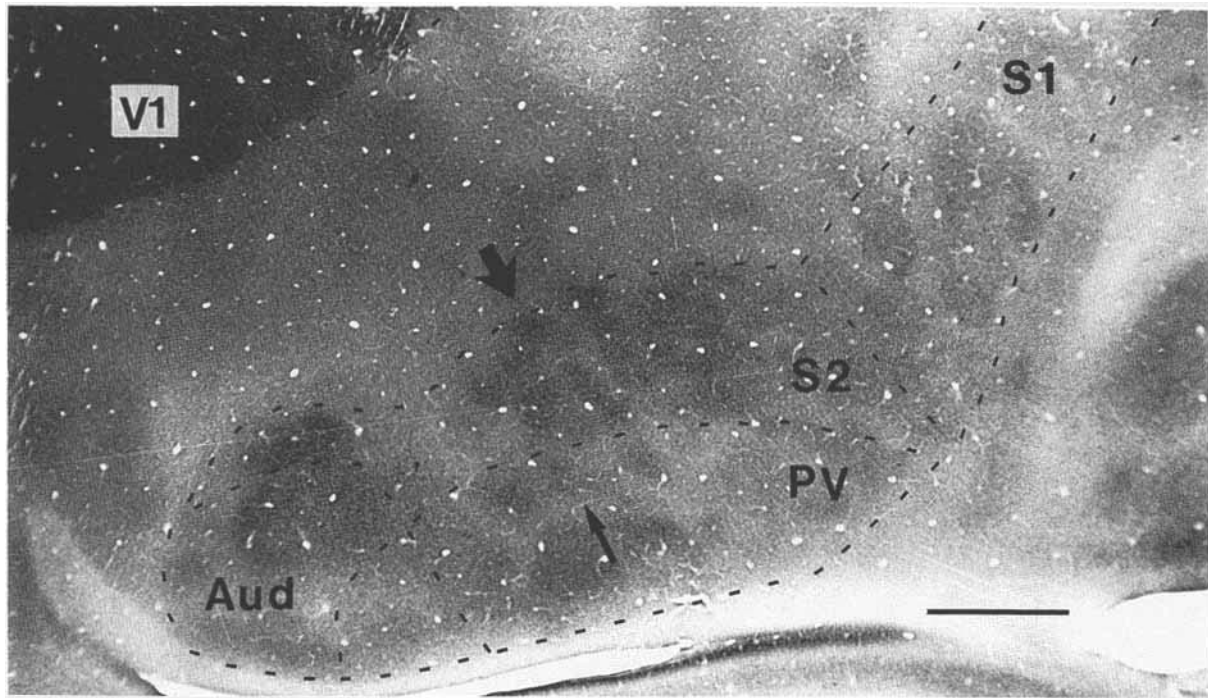


Fig. 14. Modular patterns of cytochrome oxidase staining in somatosensory cortex. The borders of S1, S2, PV, the auditory cortex, and V1 are indicated with dotted lines. Arrows point to the centers of vibrissae

fields in S2 (thick arrow) and PV (thin arrow), but receptive fields were too large to relate individual vibrissae to particular patches of staining. Scale bar = 2 mm.

connections within S1 were reciprocal. Labeled neurons and terminations were found most densely within 200–300 μm of the injection core. However, more distant labeled cells and terminals were distributed over parts of S1 both medial and lateral to the injection core, indicating that some intrinsic connections relate parts of S1 representing different body regions. In particular, intrinsic connections with the cortex medial to the injection were more widespread, suggesting that forelimb and face regions of S1 interact with trunk and hindlimb regions. In addition, a large portion of the CO-dark zone caudal to S1 that we identified as SC contained labeled neurons and terminals. These connections clearly implicate SC in the processing of somatosensory information. Likewise, the CO-light SR zone (Fig. 1) along the rostral border of S1 had dense interconnections with S1, consistent with it being a somatosensory or sensorimotor area. Finally, clusters of labeled neurons and terminals were found in S2 and PV, with PV having somewhat denser connections. Some of the labeled zones were in regions expected to represent the face and forelimb, but other regions, especially those near the S1 border, would be somatotopically mismatched with the injection site.

Similar results were apparent in a second opossum with a more lateral injection in a part of S1 largely devoted to facial vibrissae (Fig. 17). Again, there were widespread intrinsic connections, but in this case they were almost completely restricted to the parts of S1 lateral to the injection that represented the rostral face. The labeled regions of SC and SR were immediately caudal and rostral to the injection site, suggesting that these fields had somatotopic organizations that parallel those in S1. Separate foci of label in S2 and PV appeared to be in cortex that is somatotopically

congruent with the injection core in S1, whereas a more rostral focus near the S1 border suggested somatotopically mismatched connections.

In a third case (not illustrated), an injection of the fluorescent dye fluororuby was placed in medial S1 near the rostral border. As with the WGA-HRP injections, labeled neurons were found in SC, SR, S2, and PV. In addition, a few labeled cells were found scattered in the cortex dorsal to S2 and medial to S1. A second injection of diamidino yellow in SR just at the rostral border of S2 labeled neurons more laterally and medially in SR, the medial one-half of S1, across most of SR, and parts of S2 and PV. Again, a few scattered cells labeled with diamidino yellow were found elsewhere.

In summary, injections of WGA-HRP or fluororuby in S1 revealed connections between S1 and nearby regions of cortex we identified as SC, SR, S2, and PV. Thus, somatosensory cortex in opossums includes five interconnected fields.

Subdivisions of visual cortex

As part of a limited effort to identify other subdivisions of the cortex relative to the somatosensory fields, we distinguished several subdivisions of visual cortex. The CO-dark oval of the caudomedial cortex (Fig. 13A) was identified as V1 by microelectrode recordings in two opossums. Receptive fields were defined for neurons at 60 sites in one opossum and for 27 sites in a second opossum. Most of these sites were in the more accessible rostral one-half of the CO-dense oval, V1. Connections of visual cortex were examined in one opossum.

Microelectrode mapping of visual cortex. Receptive fields were referenced according to two traditional visual field coordinates. The zero horizontal meridian was located

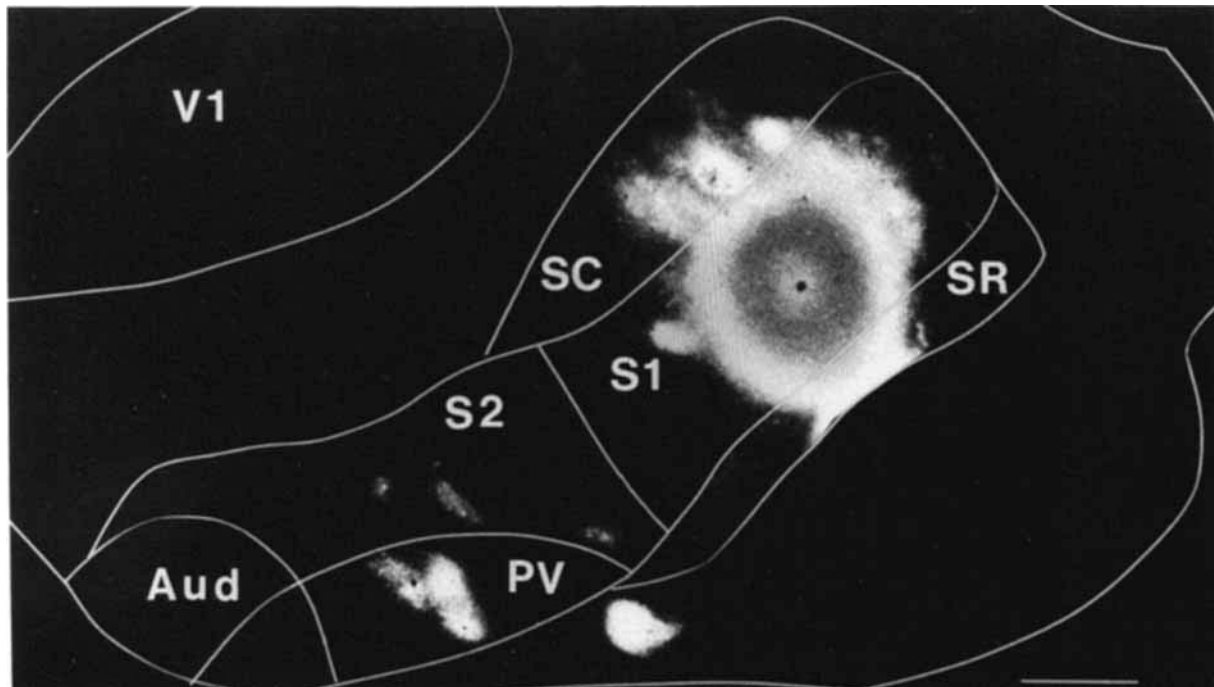


Fig. 15. Pattern of ipsilateral cortical label after an injection of wheatgerm agglutinin-horseradish peroxidase (WGA-HRP) into S1 in case 93-88. A darkfield photomicrograph shows a single unstained brain section cut parallel to the surface and processed for HRP. The glass micropipette produced a hole in the tissue that was surrounded by a dense reaction product (dark oval), representing the effective injection site, and a dense halo of transported label (white), with protrusions of

transported label both caudal (left) and rostral to S1. The rostral extent of label appears to stop sharply along a line because the cortex flattened unevenly due to the orbital dimple. Other distant foci of label were in S2 and PV. Compare with a complete reconstruction of the label illustrated in Figure 16. The dense focus of label rostral to PV corresponds to subcortical fibers, and these are not included in Figure 16. Scale bar = 2 mm.

4.5° below the projection of the optic disc of the contralateral eye into the visual field (Sousa et al., 1978), and the zero vertical meridian, or the line of decussation for retinal ganglion cells, was defined as the center of the overlap of receptive fields for neurons recorded along the lateral margin of area 17 of both hemispheres. Receptive fields for the central 15° of vision (within 15° of the intersection of the zero horizontal and vertical meridians) were 5–10° in diameter, and there was little increase in receptive field size for those located in more peripheral parts of the visual field, including those 50° into the lower visual quadrant. The cortex on the dorsolateral surface represented about 40° of horizontal vision, with at least 30° of this being the binocular cortex. The rostral one-half of area 17 in the two opossums was found to represent the contralateral lower visual hemifield in a systematic fashion, with the extreme of the lower visual quadrant (over 50° below the zero horizontal meridian) represented near the rostromedial pole of V1. The zero vertical meridian, approximately 25° nasal to the blind spot, was represented along the dorsolateral border, and a representation of some of the upper visual quadrant was revealed in the caudal one-half of area 17. Central and paracentral portions of the visual hemifield related to similar amounts of the cortex, and no parts of the retina had a highly magnified representation.

Lateral to V1, neurons responded to visual stimuli. These neurons tended to be less responsive to visual stimuli than those in V1 and had slightly larger receptive fields. Receptive fields for neurons in more rostral sites were located progressively lower in the visual field. These limited obser-

vations were consistent with the proposal that a second visual area, V2, borders lateral V1 in opossums as in other mammals.

Connections of visual cortex. In a single opossum, an injection of WGA-HRP was placed in the rostromedial V1 to reveal connections of visual cortex. Four or five evenly spaced patches of labeled neurons and terminals were located in the rostral one-half of V2, suggesting the existence of modular subdivisions. Other foci of label were in the cortex between V2 and PV, implicating at least some of this cortex in visual processing. Other foci of label were in the cortex along the rostromedial border of V1. The injection also labeled thalamic neurons in the lateral geniculate nucleus and the lateral posterior (pulvinar) nucleus.

DISCUSSION

The present results suggest that the somatosensory cortex in opossums consists of at least five interconnected fields, S1, S2, PV, SR, and SC. At least three of these fields are somatotopically organized. Part of PV contains neurons responsive to auditory stimuli, but there is also a separate auditory area, possibly A1. Visual areas V1 and V2 and a visual belt adjoining V2 are located caudomedial to the somatosensory fields. Although movements were evoked by microstimulation, no area of the cortex was clearly identified as M1.

S1 and the concept of an S1–M1 amalgam

There can be little doubt about the identity of S1 in opossums. A primary somatosensory field, S1, has been

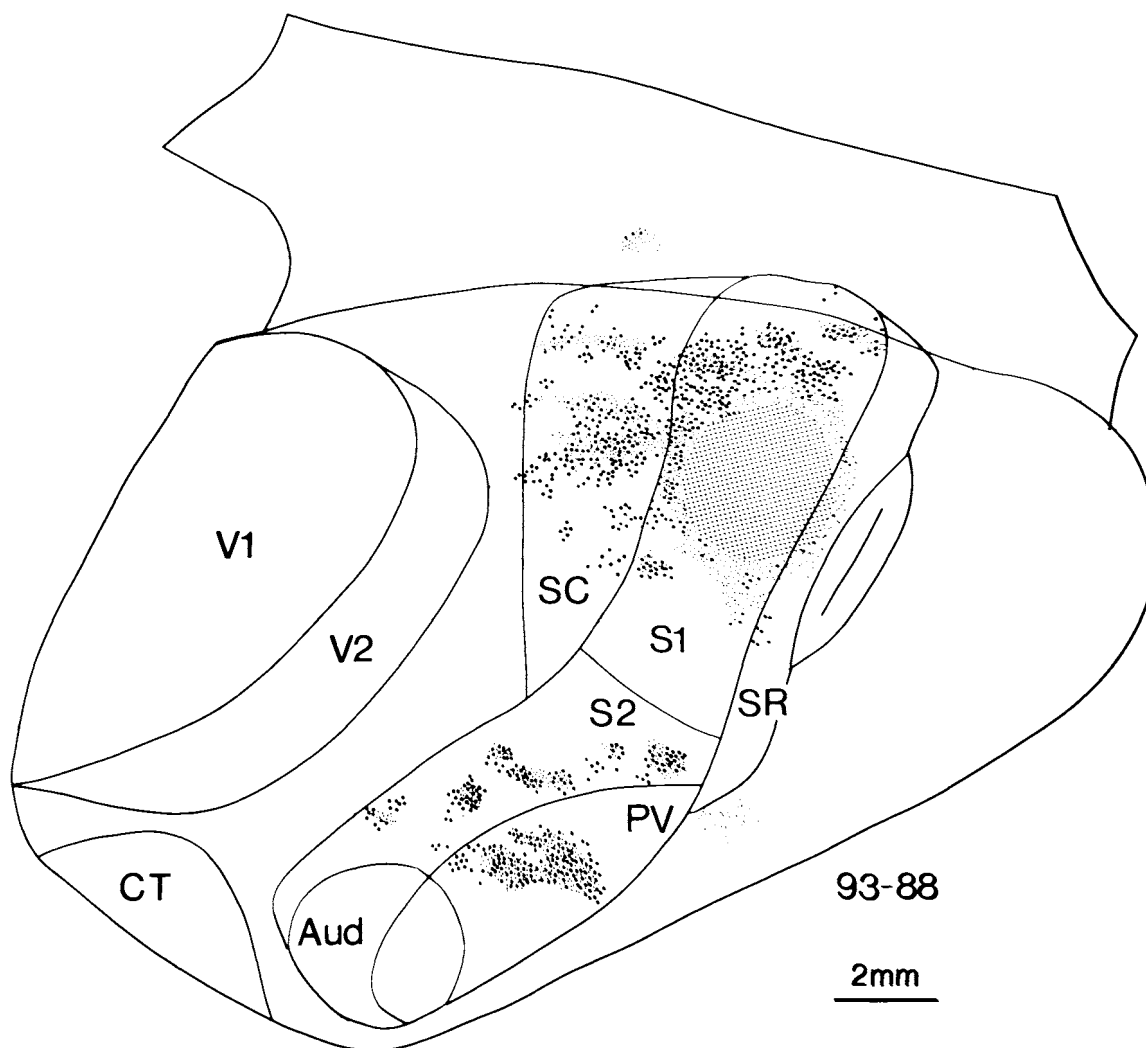


Fig. 16. Case 93-88. Ipsilateral cortical label after an injection of WGA-HRP into S1. The neocortex of the dorsolateral surface and medial wall is outlined from brain sections from the flattened cortex. Visual, auditory, and somatosensory areas are outlined (compare with

Fig. 1), as is the cortex of the orbital dimple. The effective injection site (shading) was almost completely confined to S1 (compare with Fig. 15). Transported label in fibers (fine dots) and cell bodies (large dots) was reconstructed over the depth of cortex.

located in all mammals investigated, where it can be identified by its relative position, architecture, somatotopic organization, and pattern of connections, although notable variations exist across species for all of these features (see Kaas, 1983; Johnson, 1990; Catania and Kaas, 1995; Krubitzer et al., 1995b). The primary somatosensory area as presently delineated in opossums (Fig. 1) corresponds closely with the description by Pubols et al. (1976), except that we place a second representation of the rostral face in S2 rather than in S1. As in most other mammals, S1 in opossums represents the hindlimb, trunk, forelimb, and face in a mediolateral sequence. The details of the somatotopy are consistent with the field being identified as S1. We would expect M1 to have a more complex, but parallel, somatotopic organization (see Gould et al., 1986).

The primary somatosensory area has been identified, but not as extensively studied, in several other marsupials. Recordings and electrical stimulation at the cortical surface from the South American opossum, *Didelphis azarae* (Magalhães-Castro and Saraiva, 1972), produced results similar

to those from North American opossum, *Didelphis marsupialis* (Lende, 1963a-c), in that a motor representation was superimposed on S1. However, the crude localization provided by surface electrodes and stimulation led to depictions of a larger S1-M1 that included the complete rostro-caudal length of the parietal region. Of the Australian marsupials, S1 has been mapped in the wallaby, *Macropus eugenii* (Lende, 1963a; also see Johnson, 1990), the wombat, *Vombatus ursinus* (Johnson et al., 1973; see Johnson, 1990), and the bush-tail opossum, *Trichosarus vulpecula* (Elston et al., 1993; Weller, 1993). Although the CO pattern within S1 of North American opossums has an unevenness that suggests subdivisions related to body parts (Fig. 13A), *Trichosarus* has a pattern (revealed by succinic dehydrogenase activity) that resembles the "barrel field" of S1 in rats (Dawson and Killackey, 1987) in that the arrangement of dark bands and ovals of tissue form a histological caricature of the contralateral body surface (Weller, 1993; also see Woolsey et al., 1975). Thus, the somatotopy of S1 in

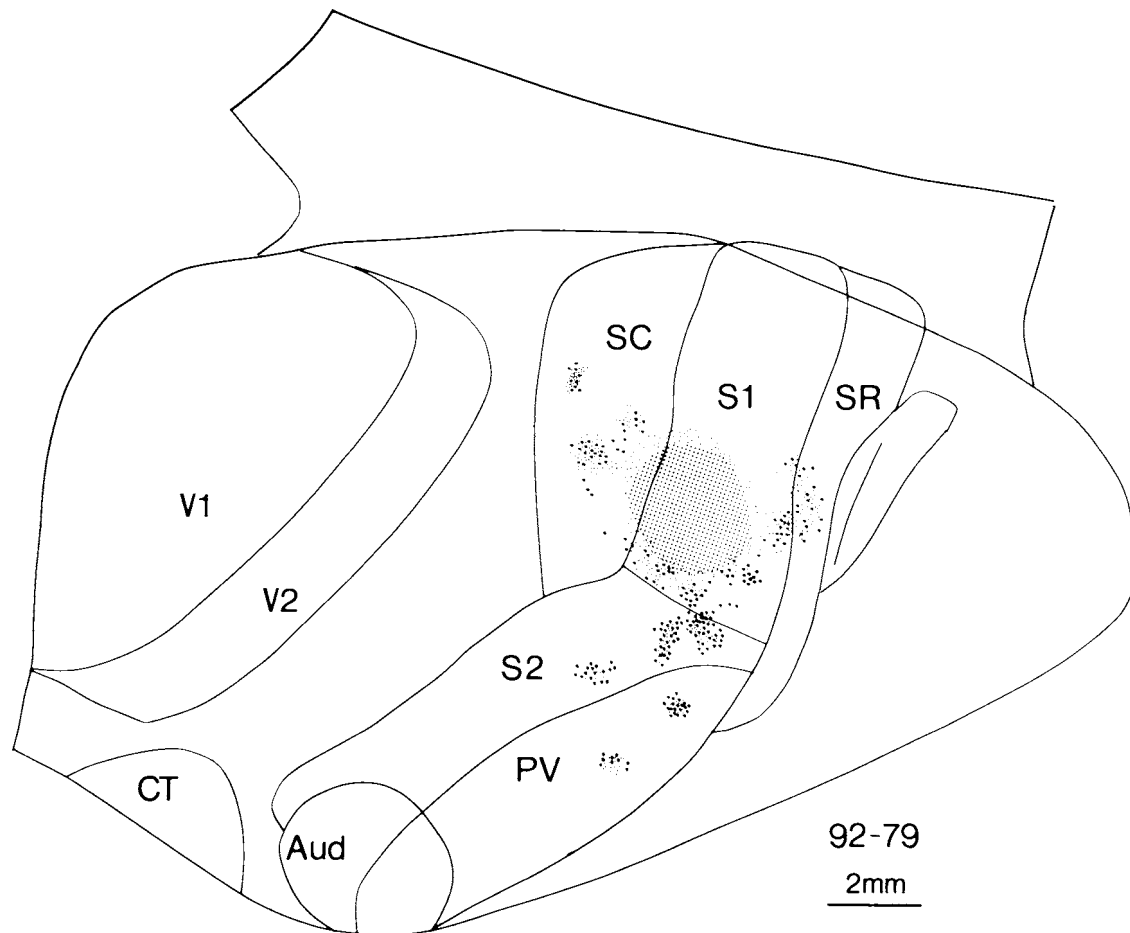


Fig. 17. Ipsilateral cortical connections of S1. Conventions as in Figure 16.

bushtail opossums can be defined rather precisely from its architecture.

S1 is within a larger cytoarchitectonic region denoted as the parietal area (PA) by Gray (1924). Although layer IV is not particularly well developed in any part of the cortex in opossums, PA has a notable granular layer (IV) and thus does not have the agranular appearance of the motor cortex (also see Walsh and Ebner, 1970). Within the larger PA, S1 is distinct along its rostral border in Nissl, myelin, and CO preparations (Fig. 13). A caudal architectonic border is not as obvious, but S1 is somewhat darker in CO preparations and more densely myelinated than the more caudal cortex.

As expected, our injections in S1 densely labeled neurons and terminals in the ventroposterior (VP) nucleus of the thalamus. S1 is the major cortical target of VP in other mammals, although other fields may receive sparser projections (see Kaas, 1988). Lesions placed in different somatotopic sectors of S1 by Pubols (1968) caused neurons in different parts of VP to degenerate in a pattern consistent with the known somatotopy of VP in opossums (Pubols and Pubols, 1966) and other mammals. Similar results were obtained by Donoghue and Ebner (1981) after injections of tracers into physiologically identified locations in S1. Donoghue and Ebner (1981) also concluded that the zone of labeled neurons in the thalamus extended caudorostrally through VP and into VL, the major thalamic relay nucleus

to primary motor cortex (Jones, 1985). This conclusion supports the concept of S1 as an S1-M1 "amalgam" (Lende, 1963a). Although VL has been identified in opossums by projections from the deep cerebellar nuclei (Walsh and Ebner, 1973), the opossum thalamus is not well differentiated (Oswaldo-Cruz and Rocha-Miranda, 1968), and a VL-VP border can be difficult to distinguish. Thus, it remains somewhat uncertain whether S1 has inputs from the motor thalamus.

The areal patterns of cortical connections of S1 in opossums resemble those of S1 of eutherian mammals. Typically, S1 projects to regions rostral and caudal to S1 and to two or more locations caudolateral to S1 (for review, see Krubitzer et al., 1993). The rostral targets include a sensorimotor transition zone that is termed 3a in primates and cats, primary motor cortex, and possibly some premotor areas. The caudal targets include a bordering zone that is differentiated into areas 1 and 2 in higher primates (see Kaas, 1983). Caudolateral connections are to S2 and PV and sometimes to additional targets. Thus, the relative positions of regions connected with S1 in opossums are similar to those reported for S1 of eutherian mammals.

The proposal of Lende (1963a) that S1 of opossums and other marsupials is actually an "amalgam" of S1 and M1 is interesting in its implications. This proposal implies more than that S1 has motor features, because we have long

Hypotheses for the Evolution of Somatosensory and Motor Cortex

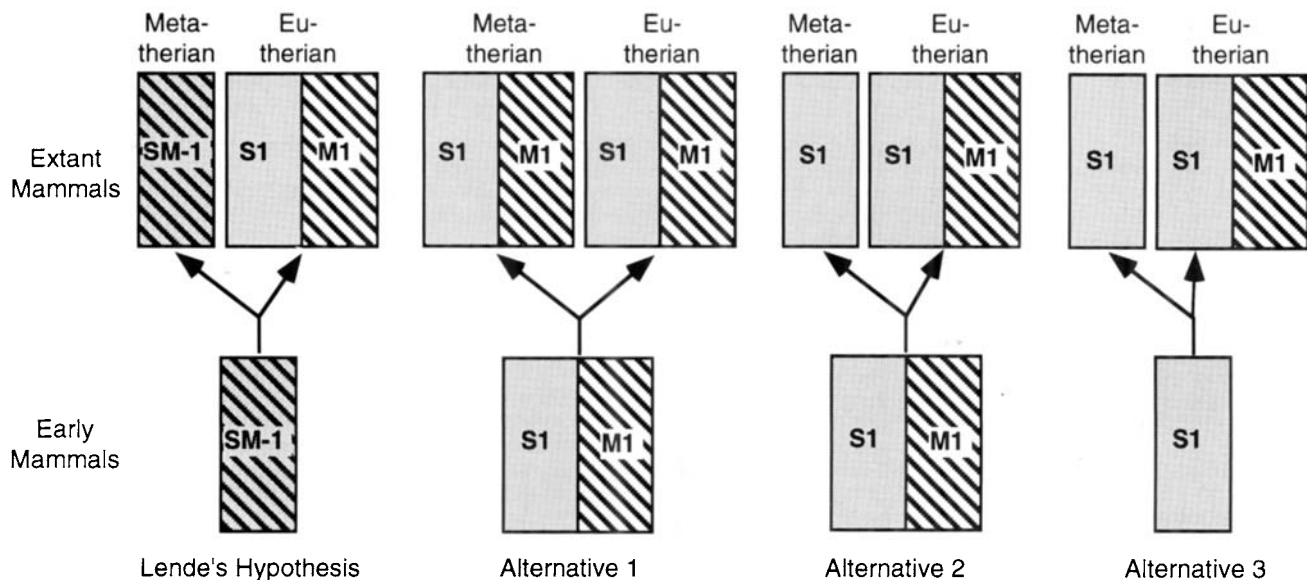


Fig. 18. Schema of four hypotheses of the evolution of the sensori-motor cortex in mammals. Lende (1963a) implied that the somatosensory and the motor cortex in the earliest mammals was amalgamated and that opossums retained this amalgam, whereas eutherian mammals evolved separate somatosensory and motor areas. Alternative hypotheses are (1) separate somatosensory and motor areas evolved from a single sensorimotor area, but the motor cortex is yet to be

described in opossums; (2) early mammals possessed separate somatosensory and motor areas, and eutherian mammals retained both areas, but opossums lost the motor area; and (3) early mammals possessed only one somatosensory area, which was retained in opossums, and eutherian mammals evolved motor areas. At present, evidence is lacking to support any single hypothesis.

known that movements can be evoked from S1 (e.g., Woolsey, 1958), and motor features could be enhanced in S1 of some mammals such as opossums. Rather, Lende's (1963a-c) papers suggest that S1 and M1 evolved in eutherian mammals from a single area that combined the traits of S1 and M1 and that this single S1-M1 area has been retained in opossums (see Fig. 18). Thus, an S1-M1 amalgam is an ancestral or plesiomorphic condition, whereas separate S1 and M1 fields are differentiated replications of a single ancestral field. According to this view, both S1 and M1 are derived homologues of the S1-M1 field (see Wiley, 1981). Such a concept of a single field differentiating into two or more distinct fields is an old, but currently supportable, explanation of interspecies differences in a number of cortical fields (e.g., Brödmann 1909; also see Kaas, 1989, 1994; Krubitzer et al., 1993).

Although the theory that S1 of opossums is actually S1 plus M1 is feasible and attractive, current evidence is not compelling. Figure 18 illustrates three alternative possibilities: (1) opossums have a motor cortex rostral to S1, but it has not been recognized; (2) early mammals had both S1 and M1, but M1 has been lost in opossums; and (3) early mammals had S1 but not M1, but later eutherian mammals evolved M1 in some way not involving a differentiation from S1. The present results do not provide a clear choice among these alternatives. Architectonically, no zone of cortex in opossums clearly resembles M1 of eutherian mammals. Microstimulation did evoke movements from S1, SR, and the cortex rostral to SR, but thresholds were higher than we would expect from the motor cortex (e.g., Stepniewska et al., 1993; however, see Sanderson et al., 1984). M1 is

expected to have some connections with S1 (e.g., Krubitzer et al., 1993), and we found rostral connections of S1 to be only with SR, an area that seems more like 3a than M1 (see below). Furthermore, a major corticospinal projection originates from M1 of eutherian mammals, but most corticospinal projections appear to originate from neurons in S1 in opossums, with a sparser number in SR and few, if any, from the cortex rostral to SR (Martin and Fisher, 1968; Nudo and Masterton, 1990). These observations argue against the possibility of an M1 in the cortex rostral to SR. However, the presence of large numbers of corticospinal neurons does not identify S1 as an S1-M1 amalgam because S1 has many corticospinal neurons in eutherian mammals (Li et al., 1990; Nudo and Masterton, 1990). In addition, corticospinal fibers in opossums (Martin and Fisher, 1968) and other marsupials (see Martin et al., 1972) terminate mainly in the medial part of the dorsal horn and in the dorsal column nuclei, locations that seem more consistent with projections from the sensory rather than the motor cortex. Similar conclusions could be made from the pattern of corticomesencephalic projections in opossums (Martin, 1968).

To complicate the situation further, at least a partially separate motor area has been reported for some marsupials (e.g., Haight and Neylon, 1979; see Johnson, 1990; Rowe, 1990), and separate or partially separate motor areas have been described for monotremes (Bohringer and Rowe, 1977; Krubitzer et al., 1995b), whereas both totally overlapping and partially separate motor areas have been claimed for edentates (for review, see Johnson, 1990). Further investigation could lead to other interpretations of cortical

organization in some of these mammals. For example, the interpretation that M1 and S1 overlap in a common region representing the hindlimb of rats (Donoghue et al., 1979) has been challenged by evidence for a separate motor representation of the hindlimb in the cortex rostral to S1 (Li et al., 1990).

Present evidence, although incomplete and far from compelling, may favor the seemingly improbable phylogenetic hypothesis that an M1–S1 amalgam is the ancestral condition and that partial and/or complete separation of these representations have evolved independently in all three major mammalian radiations, possibly more than once in each radiation (see Haight and Neylon, 1979). If M1 has evolved independently a number of times as a result of parallel evolution, the primary motor fields in different lines are not all homologous motor fields. However, M1 and S1 together would have a field homology with the M1–S1 amalgam. If M1 did evolve separately several times, other highly similar fields in different taxa may have evolved separately and need not be homologous.

S2 and PV

Our results provide strong evidence for the existence of two somatosensory areas in the cortex caudal and lateral to S1 (Fig. 1). Both of these areas represent the body from head to hindlimb in a rostrocaudal cortical sequence, and both of these areas are interconnected with S1. We term the more dorsal area S2 and the more ventral area PV. These two fields are the probable homologues of the second somatosensory area, S2, and the parietal ventral somatosensory area, PV, of other mammals.

The distinction between S2 and PV was made rather recently when it became clear in squirrels that two representations, rather than one, exist in the S2 region (Krubitzer et al., 1986). Both representations have now been described in other rodents (rats; Li et al., 1990; Fabri and Burton, 1991), megachiropteran bats (Krubitzer et al., 1993), New World monkeys (Krubitzer and Kaas, 1990), and Old World macaque monkeys (Krubitzer et al., 1995a). There is also evidence for PV in monotremes (Krubitzer et al., 1995b). The existence of S2 and PV in such distantly related mammals suggests that these fields emerged early in evolution, perhaps before or with the first mammals. In all species where S2 and PV have been described, the head is represented at or near the S1 border in both S2 and PV, and dual representations of other body parts occur in parallel away from that border. Typically, parts of PV responding to the hindlimb and trunk are also activated by auditory stimuli, as in the opossums of the present study.

The portion of cortex we define as S2 includes a representation of part of the face that was previously assigned to S1 by Pubols et al. (1976). By placing this region in S2 rather than in S1, we define S2 as a complete representation of the body and avoid the anomaly of a dual representation of the rostral face in S1. Misassignment of part of one representation to another field has been a recurring problem in deducing cortical organization (see Kaas, 1982). In particular, Adrian (1940) described a second representation of the forepaw in cats as a partial re-representation within S1, whereas Woolsey (1943) argued that it was part of a second, complete representation, S2. Because Pubols et al. (1976) placed a second representation of the rostral face of opossums in S1, a third more ventral representation of the face was placed in S2 rather than being taken as evidence for a third somatosensory field. Thus, Pubols (1977) provided

early evidence for a third area, PV, in opossums, much as Adrian (1940) provided the first evidence for S2, but the three representations of the face were not recognized as belonging to three separate areas. It appears from the illustrations of Pubols (1977) that most of the recording sites attributed to S2 were in the cortex we would assign to PV. As for much of PV in the present study, Pubols (1977) found recording sites in S2 to be responsive to both somatosensory and auditory stimuli. Earlier studies with surface electrodes did not distinguish PV from S2 in opossums, but the region termed S2 was found to overlap the region responsive to auditory stimuli (Lende, 1963b).

The somatosensory rostral and caudal areas, SR and SC

Narrow bands of cortex along both the rostral and caudal borders of S1 have dense connections with S1. Neither of these target fields was activated by light tactile stimulation in anesthetized opossums, but the cortex just rostral and just caudal to S1 in the Australian marsupial, *Trichosarus valpecula*, was found to be responsive to the stimulation of "deep" receptors (Elston et al., 1993). In *Trichosarus* and in *Didelphis*, S1 projects to both rostrally and caudally adjoining fields (Elston et al., 1993). Thus, both of these marsupials appear to have somatosensory fields along the rostral and caudal borders of S1. These fields may be activated by cutaneous stimuli via the S1 projections in awake opossums. We term these fields the rostral and caudal somatosensory areas, SR and SC.

SR and SC are within the larger region of cortex described as "parietal" by Gray (1924). Both areas have architectonic features that distinguish them from S1 (Fig. 13). Both SR and SC have an identifiable layer IV (also see Walsh and Ebner, 1970), and thus area SR does not have the cytoarchitectonic appearance of the motor cortex, M1. Thus, it seems unlikely that SR is M1. Rather, SR has an appearance more like the rostral dysgranular cortex bordering S1 in rats (e.g., Chapin and Lin, 1984) and area 3a of carnivores and primates (see Jones and Porter, 1980; Kaas, 1983). Area 3a gets inputs from a separate nucleus in the thalamus just dorsal to VP, which largely relays information from muscle spindle receptors (see Kaas, 1988). Area 3a of monkeys is highly responsive to muscle spindle activation, but it is also responsive to tactile stimuli, possibly as a result of direct inputs from S1 (area 3b). Thus, in relative position, types of activation, cytoarchitecture, and connections with S1, area SR in opossums and other marsupials resembles area 3a of cats and monkeys and the rostral dysgranular cortex of rats. The cortex in the location of SR in opossums also receives inputs from S1 and is responsive to deep receptor inputs in another marsupial, *Trichosarus* (Elston et al., 1993). The cortex in this location has been shown to receive inputs from S1 in all appropriately investigated mammals (e.g., cats: Jones and Powell, 1968; Garraghty et al., 1987; rats: Akers and Killackey, 1978; Fabri and Burton, 1991; squirrels: Krubitzer et al., 1986; tree shrews: Weller et al., 1987; bats: Krubitzer et al., 1993; monkeys: reviewed by Krubitzer and Kaas, 1990). Such similarities could have evolved independently (see above), but they suggest a common origin.

In a number of mammals, the cortex in the position of SC responds to moderately intense tactile stimulation (see Krubitzer and Calford, 1992). In monkeys, area 1 is caudal to area 3b or S1 and contains a representation of cutaneous receptors (see Kaas, 1988). For both SR and SC of opos-

sums, connections are matched for mediolateral level in S1. This suggests the existence of parallel somatotopic representations in SR and SC, such as in areas 3a and 1 of monkeys (e.g., Kaas et al., 1979). SC in opossums may correspond to a generalized somatosensory field retained from early mammals and variously differentiated in extant mammals (for related discussion, see Krubitzer et al., 1993).

Visual and auditory regions

The visual cortex in opossums consists of at least four areas. A caudomedial area of darkly stained cortex in CO preparations (Fig. 13) clearly is primary visual cortex, V1. Our recordings revealed a retinotopic organization that is typical of V1 (also see Kaas, 1980). The organization of this V1 appears to be highly similar to that reported for the South American opossum, *Didelphis marsupialis aurita* (Sousa et al., 1978). As previously reported (Coleman et al., 1977; Coleman and Clerici, 1981), V1 has dense interconnections with the lateral geniculate nucleus.

Our results also indicate that V1 or area 17 is densely interconnected with a lateral band of cortex that appears to be V2 or area 18 (Fig. 1). An injection into rostromedial V1 labeled neurons and terminals in several patches in rostral V2. Benevento and Ebner (1971) reported that the rostral striate cortex of opossums projects to the rostral peristriate cortex, whereas the caudal striate cortex projects to the caudal peristriate cortex, a pattern consistent with the concept of V2 (also see Bravo et al., 1990). The V2 region has uneven connections with the visual cortex of the other hemisphere (Granger et al., 1985; Cusick and Kaas, 1986) so that V2 contains a series of ovals of sparse interhemispheric connections separated by narrow zones with dense connections. In the mouse opossum, *Marmosa elegans*, Bravo et al. (1990) showed that V1 projections are largely confined to the ovals, demonstrating a type of modular organization that is also seen in V2 of other mammals such as tree shrews (Cusick et al., 1985).

Our injections of WGA-HRP into V1 also labeled neurons in the cortex immediately lateral to V2 and in the limbic cortex rostromedial to V1. Benevento and Ebner (1971) demonstrated V1 projections to these same regions in an earlier degeneration study, as have Bravo et al. (1990) in mouse opossums. These findings suggest that opossums have V1, V2, and at least two other visual areas and that most or all of the cortex between V2 and the somatosensory areas is visual.

A small oval of CO-dark cortex just caudal to S2 and PV responded to auditory and not somatosensory stimuli. In addition, caudal PV, and sometimes parts of S2, responded to auditory stimuli. The responsive cortex is within the larger region where Lende (1963c) recorded auditory evoked potentials with surface electrodes. Both the purely auditory and the auditory-somatosensory regions of the cortex receive inputs from the auditory thalamus. Injections placed into the cortex we now assign to the auditory and the caudal PV regions (previously considered to be S2) labeled neurons in the medial geniculate nucleus (Kudo et al., 1986). Furthermore, injections placed into the cortex electrophysiologically identified as responding to both hindlimb stimulation and auditory stimulation (considered by the investigators to be in S2 but is probably in PV) labeled neurons in both the magnocellular divisions of the medial geniculate nucleus and the hindlimb portion of the ventroposterior nucleus (Robards and Ebner, 1977).

The identity of the purely auditory region remains uncertain, but the dense input from the medial geniculate nucleus and the dark myelination of the field suggest that much or all of the region is primary auditory cortex, A1. Best frequency maps of auditory cortex have been published for two other marsupials, the native cat, *Dasyurus hallucatus* (Aitkin et al., 1986), and the bushtail opossum, *Trichosarus vulperula* (Gates and Aitkin, 1982). In both of these marsupials, a single tonotopic pattern that is characteristic of A1 (see Luethke et al., 1988) occupied nearly all of the responsive region.

Opossum cortex and the ancestral plan

The opossum cortex has long been portrayed as having few subdivisions and as having changed little over the course of mammalian evolution so that it reasonably represents the ancestral mammal (e.g., Gray, 1924; Diamond and Utley, 1963; Ebner, 1969). We now argue that the cortex of opossums contains more areas than previously described, but this conclusion does not seriously challenge the notion that the organization of cortex in opossums is indeed primitive. We have evidence for five somatosensory areas in opossums, but these areas could have emerged early in mammalian evolution, because four or five similarly placed and somatotopically organized areas have been described in a wide range of mammalian species (see above). Similarly, there is evidence for at least four visual areas, V1, V2, a limbic-visual region medial to V1, and a visual belt lateral to V2 in opossums, but other investigated mammals appear to have these same areas (for review, see Kaas and Krubitzer, 1991). Auditory cortex is less well understood, and there may be only one purely auditory field in opossums, but at least one additional auditory field has been found in other investigated mammals (see Luethke et al., 1988). The presence or absence of a primary motor area remains uncertain, but the evidence for a primary motor field has been obtained from such a wide range of mammals (for review, see Rowe, 1990; Krubitzer et al., 1995b) that it would seem to have been present in early eutherian mammals, at least. Opossums have other subdivisions of the cortex in addition to those described in the present report. A region of the frontal cortex has been described with projections from the mediodorsal nucleus (Tobias and Ebner, 1973), an architectonically defined limbic retrosplenial area has been reported (Gray, 1924), and these two are regions or areas found in other mammals. Other subdivisions of cortex certainly will be disclosed, but the most of the cortex has now been assigned to designated fields. The cortex of the first mammals most probably was very much like that of opossums in organization. In contrast, these same or similar fields occupy only a small portion of the cortex in mammals with more complex organizations, such as New and Old World monkeys, where large numbers of additional sensory and motor areas have been described. In owl monkeys, for example, there is evidence for at least five motor areas, nine somatosensory areas, 10 auditory areas, and 15 visual areas, with considerable cortex yet unassigned (Kaas, 1994). Thus, as has been long supposed (e.g. Brödmann, 1909), mammals vary considerably in number of cortical areas. However, we have increased our understanding of how mammals differ, what subdivisions are retained from a common ancestor, and what areas are new.

ACKNOWLEDGMENTS

We thank Dr. Todd Preuss for assistance on the cortical stimulation experiments. Helpful comments on the paper were provided by Drs. B.H. Pubols, Todd Preuss, Sherre Florence, and Neeraj Jain. The research was supported by NIH grant NS16446 to J.H.K. and an NRSA predoctoral training award MH10761 to P.D.B.

LITERATURE CITED

- Adrian, E.D. (1940) Double representation of the feet in the sensory cortex of the cat. *J. Physiol. (Lond.)* 98:16-18.
- Aitkin, L.M., D.R.F. Irvine, J.E. Nelson, M.M. Merzenich, and J.C. Clarey (1986) Frequency representation in the auditory midbrain and forebrain of a marsupial, the northern native cat (*Dasyurus hallucatus*). *Brain Behav. Evol.* 29:17-28.
- Akers, R.M., and H.P. Killackey (1978) Organization of corticocortical connections in the parietal cortex of the rat. *J. Comp. Neurol.* 181:513-538.
- Beck, P.D., and J.H. Kaas (1993) Evidence for three somatosensory cortical areas in opossum. *Soc. Neurosci. Abstr.* 19:764.
- Benevento, L.A., and F.F. Ebner (1971) The areas and layers of corticocortical terminations in the visual cortex of the Virginia opossum. *J. Comp. Neurol.* 141:157-190.
- Bohringer, R.C., and M.J. Rowe (1977) The organization of the sensory and motor areas of cerebral cortex in the platypus (*Ornithorhynchus anatinus*). *J. Comp. Neurol.* 174:1-14.
- Bravo, H., J. Olavarria, and S. Martinich (1990) Patterns of interhemispheric and striate-peristriate connections in visual cortex of the South American marsupial *Marmosa elegans* (mouse opossum). *Anat. Embryol.* 182:583-589.
- Brödmann, K. (1909) Vergleichende Lokalisationslehre der Grosshirnrinde in ihren Prinzipien Dargestellt auf Grund des Zellenbaues. Leipzig: Barth.
- Catania, K.C., and J.H. Kaas (1995) The organization of the somatosensory cortex of the star-nosed mole. *J. Comp. Neurol.*, 351:549-567.
- Chapin, J.K., and C.-S. Lin (1984) Mapping the body representation in the SI cortex of anesthetized and awake rats. *J. Comp. Neurol.* 229:199-213.
- Clemons, H.R., and B.E. Stein (1982) Somatosensory cortex: A "new" somatotopic representation. *Brain Res.* 235:162-168.
- Coleman, J., and W.J. Clerici (1981) Organization of thalamic projections to visual cortex in opossum. *Brain Behav. Evol.* 18:41-59.
- Coleman, J., I.T. Diamond, and J.A. Winer (1977) The visual cortex of the opossum: The retrograde transport of horseradish peroxidase to the lateral geniculate and lateral posterior nuclei. *Brain Res.* 137:233-252.
- Cusick, C.G., and Kaas, J.H. (1986) Interhemispheric connections of cortical, sensory and motor maps in primates. In F. Lepore, M. Ptito, and H.H. Jasper (eds): *Two Hemispheres—One Brain*. New York: Alan R. Liss, pp. 83-102.
- Cusick, C.G., M.G. MacAvoy, and J.H. Kaas (1985) Interhemispheric connections of cortical sensory areas in tree shrews. *J. Comp. Neurol.* 235:111-128.
- Dawson, D.R., and H.P. Killackey (1987) The organization and mutability of the forepaw and hindpaw representations in the somatosensory cortex of the neonatal rat. *J. Comp. Neurol.* 256:246-256.
- Diamond, I.T., and J.D. Utley (1963) Thalamic retrograde degeneration study of sensory cortex in opossum. *J. Comp. Neurol.* 120:129-160.
- Donoghue, J.P., and F.F. Ebner (1981) The organization of thalamic projections to the parietal cortex of the Virginia opossum. *J. Comp. Neurol.* 198:365-388.
- Donoghue, J.P., K.L. Kerman, and F.F. Ebner (1979) Evidence for two organizational plans within somatic sensory-motor cortex of the rat. *J. Comp. Neurol.* 183:647-664.
- Ebner, F.F. (1969) A comparison of primitive forebrain organization in metatherian and eutherian mammals. *Ann. NY Acad. Sci.* 167:241-257.
- Elston, G., L. Krubitzer, P. Manger, M. Calford, and T. Day (1993) The organization and connections of somatosensory cortex in the Australian marsupial, brush-tailed possum (*Trichosurus vulpecula*). *Soc. Neurosci. Abstr.* 19:764.
- Fabri, M., and H. Burton (1991) Ipsilateral cortical connections of primary somatic sensory cortex in rats. *J. Comp. Neurol.* 311:405-424.
- Gallay, F. (1979) Silver staining of myelin by means of physical development. *Neurology* 1:203-209.
- Garraghty, P.E., T.P. Pons, M.F. Hureta, and J.H. Kaas (1987) Somatosensory organization of S-III in cats. *Somatosens. Res.* 4:333-357.
- Gates, G.R., and L.M. Aitkin (1982) Auditory cortex in the marsupial possum (*Trichosurus vulpecula*). *Hear. Res.* 7:1-11.
- Gould, H.J., C.G. Cusick, T.P. Pons, and J.H. Kaas (1986) The relationship of corpus callosum connections to electrical stimulation maps of motor, supplementary motor, and the frontal eye fields in owl monkeys. *J. Comp. Neurol.* 247:297-325.
- Granger, C.G., R.B. Masterton, and K.K. Glendenning (1985) Origins of interhemispheric fibers in a callosal opossum (with comparison to callosal origins in rat). *J. Comp. Neurol.* 241:82-98.
- Gray, P.A. (1924) The cortical lamination pattern of the opossum, *Didelphys virginiana*. *J. Comp. Neurol.* 37:221-263.
- Haight, J.R., and L. Neylon (1979) The organization of neocortical projections from the ventrolateral thalamic nucleus in the brush-tailed possum, *Trichosurus vulpecula*, and the problem of motor and somatic sensory convergence within the mammalian brain. *J. Anat.* 129:673-694.
- Jerison, H.J. (1990) Fossil evidence on the evolution of the neocortex. In E.G. Jones and A. Peters (eds): *Cerebral Cortex*, vol. 8A: Comparative Structure and Evolution of Cerebral Cortex, part I. New York: Plenum Press, pp. 285-309.
- Johnson, J.I. (1990) Comparative development of somatic sensory cortex. In E.G. Jones and A. Peters (eds): *Cerebral Cortex*. New York: Plenum Press, pp. 335-449.
- Johnson, J.I., J.R. Haight, and D. Megirian (1973) Convolutions related to sensory projections in cerebral neocortex of marsupial wombats [Abstract]. *J. Anat.* 114:153.
- Jones, E.G. (1985) *The Thalamus*. New York: Plenum Press.
- Jones, E.G., and R. Porter (1980) What is area 3a. *Brain Res. Rev.* 2:1-43.
- Jones, E.G., and T.P.S. Powell (1968) The ipsilateral cortical connections of the somatic sensory areas in the cat. *Brain Res.* 9:71-94.
- Kaas, J.H. (1970) Cortical visual area I and II in the hedgehog: The relation between evoked potential maps and architectonic subdivisions. *J. Neurophysiol.* 33:595-615.
- Kaas, J.H. (1980) A comparative survey of visual cortex organization in mammals. In S.O.E. Ebberson (ed): *Comparative Neurology of the Telencephalon*. New York: Plenum Press, pp. 483-502.
- Kaas, J.H. (1982) The segregation of function in the nervous system: Why do the sensory systems have so many subdivisions? *Contrib. Sens. Physiol.* 7:201-240.
- Kaas, J.H. (1983) What, if anything, is SI? Organization of first somatosensory area of cortex. *Physiol. Rev.* 63:206-230.
- Kaas, J.H. (1987) The organization and evolution of neocortex. In S. Wise (ed): *Higher Brain Function: Recent Exploration of the Brain's Emergent Properties*. New York: John Wiley & Sons, pp. 347-377.
- Kaas, J.H. (1988) How the somatosensory thalamus is subdivided and interconnected with areas of somatosensory cortex in monkeys. In M. Bentivoglio and R. Spreafico (eds): *Cellular Thalamic Mechanisms*. Amsterdam: Elsevier Science Publications, pp. 143-150.
- Kaas, J.H. (1989) The evolution of complex sensory systems in mammals. *J. Exp. Biol.* 146:165-176.
- Kaas, J.H. (1994) The organization of sensory and motor cortex in owl monkeys. In J.F. Baer, R.E. Weller, and I. Kakoma, (eds): *Aotus: The Owl Monkey*. Orlando: Academic Press, pp. 331-351.
- Kaas, J.H., and L.A. Krubitzer (1991) The organization of extrastriate visual cortex. In B. Dreher and S.R. Robinson (eds): *Neuroanatomy of Visual Pathways and Their Development: Vision and Visual Dysfunction*. London: Macmillan, pp. 302-323.
- Kaas, J.H., and L.A. Krubitzer (1992) Area 17 lesions deactivate area MT in owl monkeys. *Vis. Neurosci.* 9:399-407.
- Kaas, J.H., W.C. Hall, and I.T. Diamond (1970) Cortical visual area I and II in the hedgehog: The relation between evoked potential maps and architectonic subdivisions. *J. Neurophysiol.* 33:595-615.
- Kaas, J.H., R.J. Nelson, M. Sur, C.-S. Lin., and M.M. Merzenich (1979) Multiple representations of the body within the primary somatosensory cortex of primates. *Science* 204:521-523.
- Krubitzer, L.A., and M.B. Calford (1992) Five topographically organized fields in the somatosensory cortex of the flying fox. *J. Comp. Neurol.* 317:1-30.
- Krubitzer, L.A., and J.H. Kaas (1989) Cortical integration of parallel pathways in the visual system of primates. *Brain Res.* 478:161-165.
- Krubitzer, L.A., and J.H. Kaas (1990) The organization and connections of somatosensory cortex in marmosets. *J. Neurosci.* 10:952-974.

- Krubitzer, L.A., M.A. Sesma and J.H. Kaas (1986) Microelectrode maps, myeloarchitecture, and cortical connections of three somatotopically organized representations of the body surface in the parietal cortex of squirrels. *J. Comp. Neurol.* 250:403-430.
- Krubitzer, L.A., M.B. Calford, and L.M. Schmid (1993) Connections of somatosensory cortex in megachiropteran bats: The evolution of cortical fields in mammals. *J. Comp. Neurol.* 327:473-506.
- Krubitzer, L.A., J. Clarey, R. Tweedale, G. Elston, and M. Calford (1995a) The organization of lateral somatosensory cortex in macaque monkey: Where is SII? *J. Neurosci.* 15:3821-3839.
- Krubitzer, L.A., P.R. Manger, J.D. Pettigrew, and M. Calford (1995b) The organization of somatosensory cortex in monotremes: In search of the prototypical plan. *J. Comp. Neurol.* 351:261-306.
- Kudo, M., K.K. Glendenning, S.B. Frost, and R.B. Masterton (1986) Origin of mammalian thalamocortical projections. I. Telencephalic projections of the medial geniculate body in the opossum (*Didelphis virginiana*). *J. Comp. Neurol.* 245:176-197.
- Lende, R.A. (1963a) Cerebral cortex: A sensorimotor amalgam in the marsupial. *Science* 141:730-732.
- Lende, R.A. (1963b) Sensory representation in the cerebral cortex of the opossum (*Didelphis virginiana*). *J. Comp. Neurol.* 121:395-403.
- Lende, R.A. (1963c) Motor representation in the cerebral cortex of the opossum (*Didelphis virginiana*). *J. Comp. Neurol.* 121:405-415.
- Li, X.-G., S.L. Florence, and J.H. Kaas (1990) Areal distribution of cortical neurons projecting to different levels of the caudal brain stem and spinal cord in rats. *Somatosens. Res.* 7:315-335.
- Luethke, L.E., L.A. Krubitzer, and J.H. Kaas (1988) Cortical connections of electrophysiologically and architectonically defined subdivisions of auditory cortex in squirrels. *J. Comp. Neurol.* 268:181-203.
- Magalhães-Castro, B., and P.E.S. Saraiva (1972) Sensory and motor representation in the cerebral cortex of the marsupial *Didelphis azarae* azara. *Brain Res.* 34:291-299.
- Martin, G.F. (1968) The pattern of neocortical projections to the mesencephalon of the opossum, *Didelphis virginiana*. *Brain Res.* 11:593-610.
- Martin, G.F., and A.M. Fisher (1968) A further evaluation of the origin, course, and termination of the opossum corticospinal tract. *J. Neurol. Sci.* 7:177-189.
- Martin, G.F., P. Megirian, and J.B. Conner (1972) The origin, course and termination of the corticospinal tracts of the Tasmanian potoroo (*Potoroos apicalis*). *J. Anat.* 111:263-281.
- Martinich, S., M.G.P. Rosa, and C.E. Rocha-Miranda (1990) Patterns of cytochrome oxidase activity in the visual cortex of a South American opossum (*Didelphis marsupialis aurita*). *Brazil. J. Med. Biol. Res.* 23:883-887.
- Mesulam, M.M. (1978) Tetramethylbenzidine for horseradish peroxidase neurohistochemistry. *J. Histochem. Cytochem.* 26:160-177.
- Mesulam, M.M. (1982) Principles of horseradish peroxidase neurohistochemistry and their applications for tracing neural pathways—Axonal transport, enzyme histochemistry and light microscopic analysis. In M.M. Mesulam (ed.): *Tracing Neural Connections With Horseradish Peroxidase*. Chichester: John Wiley & Sons, pp. 1-151.
- Nudo, R.J., and R.B. Masterton (1990) Descending pathways to the special card, III: Sites of origin of the corticospinal tract. *J. Comp. Neurol.* 296:559-583.
- Oswaldo-Cruz, E., and C.E. Rocha-Miranda (1968) The Brain of the Opossum (*Didelphis marsupialis*). A Cytoarchitectonic Atlas in Stereotaxic Coordinates. Rio de Janeiro: Instituto de Biofísica.
- Pubols, B.H. (1968) Retrograde degeneration study of somatic sensory thalamocortical connections in brain of Virginia opossum. *Brain Res.* 7:232-251.
- Pubols, B.H. (1977) The second somatic sensory area of the opossum neocortex. *J. Comp. Neurol.* 174:71-78.
- Pubols, B.H., and L.M. Pubols (1966) Somatic sensory representation in the thalamic ventrobasal complex of the Virginia opossum. *J. Comp. Neurol.* 127:19-34.
- Pubols, B.H., L.M. Pubols, D.J. Dipette, and J.C. Sheely (1976) Opossum somatic sensory cortex: A microelectrode mapping study. *J. Comp. Neurol.* 165:229-246.
- Robards, M.J., and F.F. Ebner (1977) Thalamic neurons projecting to the second somesthetic area (SII) of the cortex of the Virginia opossum. *Soc. Neurosci. Abstr.* 3:490.
- Rowe, M. (1990) Organization of the cerebral cortex in monotremes and marsupials. In E.G. Jones and A. Peters (eds): *Cerebral Cortex*. New York: Plenum Press, pp. 253-334.
- Rowe, T. (1993) Phylogenetic systematics and the early history of mammals. In F.S. Szalary, M.J. Novacek, and M.C. McKenna (eds): *Mammal Phylogeny: Mesozoic Differentiation, Multituber-Culates, Monotremes, Early Therians, and Marsupials*. New York: Springer-Verlag, pp. 129-145.
- Sanderson, K.J., W. Welker, and G.M. Shambels (1984) Reevaluation of motor cortex and sensorimotor overlap in cerebral cortex of albino rats. *Brain Res.* 292:251-260.
- Sousa, A.P.B., R. Gattase, and E. Oswaldo-Cruz (1978) The projection of the opossum's visual field on the cerebral cortex. *J. Comp. Neurol.* 177:569-588.
- Stepniewska, I., T.M. Preuss, and J.H. Kaas (1993) Architectonics, somatotopic organization, and ipsilateral cortical connections of the primary motor area (MI) of owl monkeys. *J. Comp. Neurol.* 330:238-271.
- Tobias, T.J., and F.F. Ebner (1973) Thalamocortical projections from the mediodorsal nucleus in the Virginia opossum. *Brain Res.* 52:79-96.
- Walsh, T.M., and F.F. Ebner (1970) The cytoarchitecture of somatic sensory motor cortex in the opossum (*Didelphis marsupialis virginiana*): A Golgi Study. *J. Anat.* 107:1-18.
- Walsh, T.M., and F.F. Ebner (1973) Distribution of cerebellar and somatic lemniscal projections to the ventral complex of the Virginia opossum. *J. Comp. Neurol.* 147:427-446.
- Welker, W. (1990) Why does cerebral cortex fissure and fold? A review of determinants of gyri and sulci. In E.G. Jones and A. Peters (eds): *Cerebral Cortex*, vol. 8B: *Comparative Structural and Evolution of Cerebral Cortex*, part II. New York: Plenum Press, pp. 1-136.
- Weller, W.L. (1993) SmI cortical barrels in an Australian marsupial, *Trichosurus vulpecula* (brush-tailed possum): Structural organization, patterned distribution, and somatotopic relationships. *J. Comp. Neurol.* 337:471-492.
- Weller, R.E., M. Sur, and J.H. Kaas (1987) Callosal and ipsilateral cortical connections of the body surface representations in S-I and S-II of tree shrews. *Somatosens. Res.* 5:107-133.
- Wiley, E.O. (1981) *Phylogenetics*. New York: John Wiley & Sons.
- Wong-Riley, M. (1979) Changes in the visual system of monocularly sutured or enucleated cats demonstrable with cytochrome oxidase histochemistry. *Brain Res.* 171:11-28.
- Woolsey, C.N. (1943) "Second" somatic receiving areas in the cerebral cortex of cat, dog and monkey. *J. Neurophysiol.* 32:170-183.
- Woolsey, C.N. (1958) Organization of somatic sensory and motor areas of the cerebral cortex. In H. Harlow and C. Woolsey (eds): *Biological and Biochemical Bases of Behavior*. Madison: University of Wisconsin Press, pp. 63-81.
- Woolsey, T.A., C. Welker, and R.H. Schwartz (1975) Comparative anatomical studies of the SmI face cortex with special reference to the occurrence of "barrels" in layer IV. *J. Comp. Neurol.* 164:79-94.
- Zilles, K. (1990) Anatomy of the neocortex: Cytoarchitecture and myeloarchitecture. In B. Kolb and R.C. Tees (eds): *The Cerebral Cortex of the Rat*. Cambridge: MIT Press, pp. 77-112.